E-ISSN: 3139-0668
Volume: I, Issue: II

https://kalingauniversity.ac.in/International-Journal-of-Research-and-Studies

INTERNATIONAL JOURNAL OF RESEARCH AND STUDIES

Month/Year : April-June, 2026 | DOI: https://doi.org/10.5281/zenodo.23611492

RECEIVED 27.04.2026

ACCEPTED 02.05.2026

PUBLISHED 30-06-2026

PEPTIDE-BASED STRATEGIES FOR DIAGNOSIS AND THERAPY OF LUNG CANCER: MECHANISMS, DELIVERY SYSTEMS, AND CLINICAL TRANSLATION

Aditya Nishad - Research Scholar

Department of Pharmacy, Kalinga University, Raipur, Chhattisgarh

Sandip prasad Tiwari - Assistant Professor

Faculty of Pharmacy, Kalinga University, Raipur, Chhattisgarh

ABSTRACT:

One of the most prevalent and dangerous diseases in the world today is lung cancer, with non- small cell lung cancer, with non-small cell lung cancer (NSCLC) making up the majority of cases. It is challenging to test novel therapies because conventional ones, like as radiation, chemotherapy and surgery are still insufficient particularly in later stages. Because of their excellent specificity, biocompatible and adaptability peptide-based approaches have become viable tools for diagnosis and treatment. These include cell penetrating peptides (CPPs) that react to Ph, hypoxia or enzymes, tumor-homing ligands that enhance selective delivery and peptides that target tumors such as integrin and EGFR-binding sequences. While peptide-vulnerable modulators against PD-1/ PD-L1 and CTLA-4 increase immunotherapy op operations, peptide-medication conjugates (PDCs) and peptide obstructions of carcinogenic pathways (KRAS, EGFR, c-MET) provide focused remedial choices. Additionally, stable peptides or peptidomimetics improve pharmacokinetic and resistance to decline, while peptide vaccination and neoantigen-based strategies promote tumor-specific susceptible responses. Compared to antibody-based tracers, individual procedures such peptide-based imaging tests for PET, SPECT, MRI, and near IR offer improved tumor visibility. Despite obstacle such subtype diversity, insecurity, and a lack of knowledge on tone-assembly, peptide-based nanostructured and delivery methods establish important implicit to improve lung cancer operation. These tactics often combine enhanced bioavailability, sophisticated diagnostic and flawless remediation. This allows for the restatement of clinical result and the justification of medication use in the treatment of lung cancer.

Keywords: Lung Cancer, Non-Small Cell Lung Cancer (NSCLC), Peptide-Based Therapy, Cell-Penetrating Peptides, Targeted Drug Delivery, Peptide–Drug Conjugates, Cancer Immunotherapy, Tumor Imaging, Peptidomimetics,

1. INTRODUTION

In clinicopathological condition, lungs tumors are connected to a wide range of etiologies, and they often lead to from altered secretory and basal cell multiplying capability in lungs (Sainz de Aja et al., 2021). Global Cancer Statistic 2022(GLOBACAN) forecast that there were 9.7 million cancer related death and 20 million new cases in 2022. Lung cancer accounted for 12.4% of all cancer cases, making is the most often diagnosed cases among these (Bray et al., 2024). The World Health Organization (WHO) uses histology to categorize lung cancer, emphasizing the significant of diagnosis and the molecular pathophysiology of lung malignancies. Non-small cell lung cancer (NSCLC) constitutes 85% of lung cancer cases, while small cell lung cancer (SCLC) represents 15%.Adenocarcinoma (ADC) squamous cell carcinoma (SCC), and large cell carcinoma (LCC) are other subtype of NSCLC(Inamura, 2017)(Ramos-Paradas et al., 2022). SCLC is classified into two subtype: pure SCLC with neuroendocrine markers and combined SCLC(Raso et al., 2021). ADC is regarded as a more prevalent type of lung cancer compared to others. Smoking, using tobacco, being around pollutants both indoors and outdoors, working in a dangerous job, and having a family history of lung cancer are all risk factors for different types of the disease.(Singh et al., 2021). Surgery, endobronchial treatment, adjuvant therapy, radiation therapy, chemotherapy targeted agents, and immunotherapy depending on the cancer's stage(*D.Q.A.T.E. Board, Non-Small Cell Lung Cancer Treatment (PDQ®): Health Professional Version, in: PDQ Cancer Information Summaries, National Cancer Institute (US), Bethesda (MD).*, n.d.). Approximately 60% of lung adenocarcinomas in the western population and 80% in Asian population have targetable genetic mutations(Friedlaender et al., 2024). IN the lungs the three most prevalent genetic causes of adenocarcinoma are Kirsten rat sarcoma viral oncogene homolog (KRAS), Anaplastic lymphoma kinase (ALK), and epidermal growth factor receptor (EGFR). These mutation often activate in a way that is mutually exclusive, resulting in different disease grouping. Targeted therapies have been developed for lung malignancies with EGFR and ALK mutations; however, KRAS-mutant lung cancer remained untreatable for decades until recent advancements. (Prior et al., 2020a). Although being the first oncogenes identified in the area of cancer research about forty years ago, no drug has been created to treat rat sarcoma (RAS). I believe that the battle against the RAS mutation is going well. Studies show that 19% of cancer patients have a RAS mutation, meaning that anti-RAS therapy is required for the 3.4 million new cases of cancer that are discovered globally each year.(Prior et al., 2020a) Among the RAS GTPase family members, three iso- forms have been identified: Harvey rat sarcoma viral oncogene homolog (HRAS), neuroblastoma rat sarcoma viral oncogene homolog (NRAS), and KRAS . These RAS oncogenes have the same molecular structures, but they are expressed in different ways. This helps explain why they are found in different types of cancer in people. KRAS is the most

frequently mutated RAS oncogene, representing around 76% of RAS-mutated malignancies, succeeded by NRAS (17%) and HRAS (7%). KRAS mutations are very common in cancers like pancreatic adenocarcinoma (88%), lung adenocarcinoma (31%), and colorectal cancers (45–50%). HRAS mutations are most commonly found in pheochromocytoma, paraganglioma, thymoma, and head and neck squamous cell carcinoma. NRAS mutations, on the other hand, are most commonly found in melanomas, acute myeloid leukemia, and thyroid cancer(Prior et al., 2020a). More than 85% of lung cancer cases are non-small cell lung cancer (NSCLC), which is the most common type of lung cancer. (Z. Yang & Tam, 2016). The lack of timely diagnosis (Stage III and IV) is considered as the main reason for the survival rate. Traditional treatments for NCSLC, including surgery, chemotherapy and radiation, are insufficient to reduce the high mortality rate(Barker et al., 2001)(Doebele, 2019). Peptides are short sequences of amino acids that binds specifically to molecular target. This ability makes them useful as therapeutic and diagnostic tools for cancer therapy [16]. Peptides offer a lot of potential for the improvement of lung cancer therapies in multiple ways such as peptide drug conjugates(PDCs),(Pikor et al., 2013) Peptide vaccines, peptide drugs, peptide-nanoparticles conjugates peptide inhibitors. DTX-P7 is a heptapeptide–Docetaxel conjugate, target the Hsp90 cell surface protein and kills dormant cancer stem-like cells (CSLCs). The efficacy for the treatment of NSCLC has been demonstrated(Karankar et al., 2025). Cilengitide, which is cyclic peptide is and integrin v3 and v5 antagonist it is in the clinical trial and show inhibition of NSCLC in the combination therapy(Sainz de Aja et al., 2021). In the phase 1 clinical trial the IDO peptide vaccine showing sustain long term responses with reduced toxicity compared to chemo-immunotherapy in stage III-IV patients(Drakopanagiotakis et al., 2024). We showed that, the use of peptide CIGB-522 together with the established therapy (cisplatin) improved efficiency of the chemotherapeutic drugs on lung cancer model, as proved by Yolanda et al (Bray et al., 2024).Through a number of mechanisms, peptides are highly specific to lung cancer cells. They been specifically designed to bind to lung cancer cells by binding to the receptors that are overexpressed in lung cancer such as EGFR or VEGFR. Tumor-homing sequence can effectively increase the accuracy with which peptides bind to the cancer cell by targeting the integrins that are highly overexpressed in tumor microenvironment (TME))(M. Y. Chen et al., 2022)(X. Wang et al., 2025). Although peptides have a lot of potential for lung cancer treatment the majority of peptide drugs are designed based on different lung cancer subtypes. In NSCLC, peptide targets are mainly focused on EGFR and PD-L1 in SCLC, bombesin and DLL3 are typical targets because different subtypes are different(Stephens et al., 2021). Co-administration of peptide with immunotherapy and chemotherapy may be effective and expand the clinical applications among subtype(Milewska et al., 2024).Various nanostructures, possessing improved properties as compared to conventional non biological materials can be formed by peptides because they exhibit a range of physico-chemical properties .Peptide self-

assembled nanostructures include but are not limited to nanoparticles, nanotubes, nanofibers and hydrogels, and have been widely studied for drug delivery applications(T. Fan et al., 2017) Cancer drugs should possess the capacity to surpass body's defence mechanisms, enter the desired site of action, and accumulate to an appreciable concentration . In this regard, self-assembled peptide materials have been shown to display a tuneable pharmacokinetic profile and drug targeting specificity. However, the advantages of peptide materials lead to think about their potential to delivery anti-cancer drugs. But there are many difficulties, such as the use of very short peptide sequence the cancer targeting inappropriate delivery or cargo release and stimulus responsive release. (V. B. Kumar, 2025). These can be partly attributed to our limited understanding of the formation and properties of currently-available peptide-based materials, which impedes their further improvement, design and optimization(T. Li et al., 2022). A combination of computational methods was used to investigate the cyclo-di histidine (cyclo-HH) peptide in the absence and presence of Zn 2+ ions. Although numerous reviews on peptides and the self-assembly properties of nanoscale materials have been published, the majority focus on tissue engineering and drug delivery rather than on smart or targeted drug delivery applications(V. B. Kumar, 2025).

2. Peptide Classes Relevant to Lung Cancer

2.1. Tumour Targeting motifs

2.1.1. EGFR-Targeting Peptides

The EGFR is a 170 kDa single-chain transmembrane glycoprotein that is a member of the ERBB family of receptor tyrosine kinases. It is also known as ERBB-1 (Erythroblastic Leukemia Viral Oncogene Homologue-1) or HER-1 (Human Epidermal Growth Factor Receptor-1). EGFR has three domains: hydrophobic transmembrane domain linked to dimerization interactions between receptors; an intracellular tyrosine kinase domain that phosphorylates substrates; and an extracellular EGFR-binding domain (sEGFR) that binds ligands that activate EGFR.[23][24]. Conformational modifications brought about by ligand binding reveal a dimerization arm that promotes the formation of homodimers and heterodimers with other members of the ERBB family. The catalytic area, hinge site, activation loop, and kinase specificity pocket are the four essential structural locations found in the gap between the two lobes that make up the kinase domain, which is activated by this dimerization(Hubbard, 1997),(Guy et al., 1994),(Rodrigues Toledo et al., 2024). A number of EGFR-targeting peptides have been found and are being explored as possible therapeutic and diagnostic tools. Phage display libraries, computer-aided design, experimental synthesis, and natural sources can all yield EGFR

peptides. GE11 is the most extensively studied peptide.(Rodrigues Toledo et al., 2024). Phage display screening is used to identify the dodecapeptide GE11. It has a strong affinity and specificity for EGFR. Its dissociation constant (Kd) for binding to EGFR is 22 nM, which is much greater than that of non-specific binding to proteins like bovine serum albumin but lower than that of the natural ligand EGF (Kd = 2 nM). GE11's smaller size and binding to a single EGFR region, as opposed to EGF's interaction with numerous domains, are the reasons for this reduced affinity when compared to EGF.(Genta et al., 2017). Molecular docking modeling and molecular dynamics were used to determine the binding location between GE11 and sEGFR. These researchers claim that GE11 functions as an inhibitor. GE11 and sEGFR have their ideal conformation in domains II and IV, which may limit dimer formation by blocking the dimerization arm's exposure(Decker et al., 2022)(Hailing et al., 2022).

2.1.2. Integrin *αvβ*3-Targeting Peptides

integrins are a family of transmembrane cell surface receptors that mediate cell–matrix and cell–cell interactions(Ruoslahti & Pierschbacher, 1986).These heterodimeric glycoproteins consist of noncovalently connected α and β subunits. In mammals, 18 α and 8 β subunits combine to form at least 24 distinct integrin heterodimers(Cai & Chen, 2008). Each integrin subunit has a large extracellular domain (750–1000 residues), a single transmembrane domain, and a short cytoplasmatic domain(Hood & Cheresh, 2002)(Alghisi & Rüegg, 2006) Integrins bind to specific endogenous ligands, including cell surface counter-receptors, soluble ligands, and extracellular matrix (ECM) proteins(Eble & Haier, 2006). Eight integrin heterodimers can recognize RGD-containing peptides Ligand binding triggers interactions between the cytoplasmatic domain, intracellular proteins, and cytoskeletal filaments, initiating signalling v pathways through the Src kinases which are activated by FAK phosphorylation(Ruoslahti & Pierschbacher, 1986). Integrin αVβ3, also known as the vitronectin receptor, is of particular interest in cancer research. While αVβ3 is expressed at low levels or is undetectable in most healthy adult epithelia, it is overexpressed in various tumour cells (breast, lung, glioblastoma, prostate, and melanoma), the tumour-associated vasculature, and invasive tumour fronts(Guo & Giancotti, 2004). Interactions between αVβ3 and its ECM ligands activate intracellular signalling pathways like PI3K/Akt and ERK/MAPK, promoting cancer progression. The engagement of αVβ3 with the ECM also facilitates crosstalk with growth factor receptors and proteases to further enhance tumour cell survival and motility(C. Kumar, 2005). Therefore, targeting EGFR and/or integrin αVβ3 with specific inhibitors or antibodies/small molecules can disrupt these signalling pathways, inhibiting tumour growth and spread.

2.2. cell penetration peptide (CPPS)

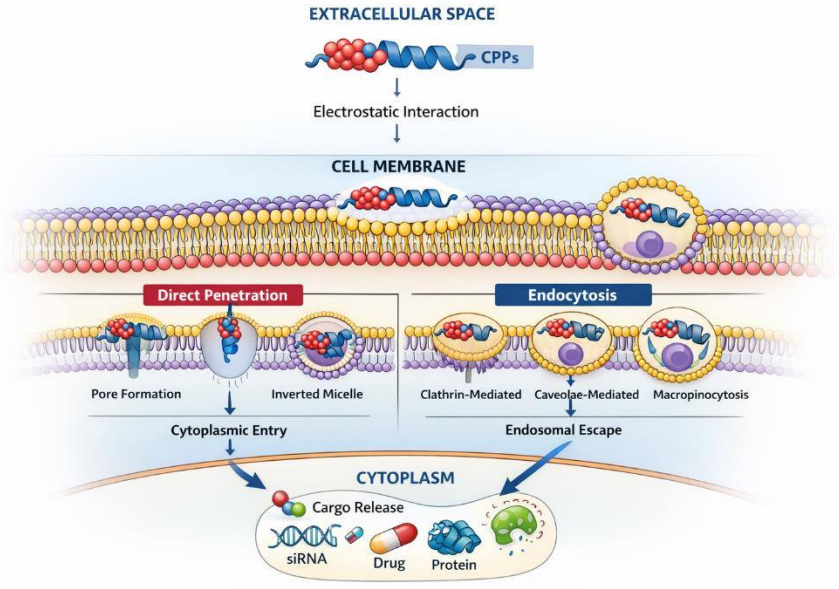


Figure 1cell penetration peptide (CPPs)

CPPs are up to 40 amino acids long peptides, with the ability to gain access to the cell interior by means of different mechanisms, and/or with the capacity to promote intracellular effects by themselves or by the delivered bioactive cargoes(Milletti, 2012)(Zorko et al., 2022).There are some Strategies for targeting different tumour cells they are.

2.2.1. pH sensitive methods

It is known that cancer cells require a high amount of energy frequently supplied by an increased level of anaerobic glycolysis. This is accompanied by a high production of lactate and protons. Therefore, the surrounding tumour microenvironment is usually acidic, which assists in tumour growth, metastasis and drug resistance This can be used to target tumour tissue by pH-responsive peptides which can be coupled to a CPP and an anticancer cargo or to a nanocarrier(Zorko et al., 2022). particular interest are so called pH low insertion peptides (pHLIPs), introduced and presented by Han and coauthors(Han et al., 2020).

2.2.2. Hypoxia responding approaches

Another characteristic of the tumour microenvironmental that has been utilize to target cancerous cells is hypoxia. Tat fused to a portion of the oxygen dependent breakdown of tissue region of the hypoxia-inducible factor-1 a protein, show that whereas degradation takes place in the immediate microenvironment of normal cells, leaving the latter untouched, such fusion construct are persistent in hypoxia condition and may penetrate cancer cells. This method may be used with a PH sensitive CPP(Harada et al., 2002).

2.2.3. Enzyme-triggering activation

Another strategy for target specific delivery, especially to cancer tissue, is the creation of activable CPP. Different types of enzymes that are expressed by healthy tissue are extracellular deposited on the membranes of tumour cells. Matrix metalloprotease are a class of endopeptidase that may breakdown protein structure are crucial for tumour invansion and metastasis and are found in the extracellular matrix. The R.Tsien group was the first adopt the term a CPPs by using a negatively charged peptide domain for intramolecular shielding, they prevented positively charged polyarginine CPPs(Jiang et al., 2004). Positively charged polyarginine CPPs were inhibited by them by using a negatively charged peptide for intramolecular shielding domain. The shielding domain and CPP are linked by a appropriate peptide linker that metalloproteases may cleave. This reactivity the cpp's capacity to penetrate cells. By proton in the microenvironment around tumour cells, and the aCPP may transport the addition to its enormous potential for targeted administration of anti-cancer medications(Soler et al., 2016).

2.2.4. Tumour homing ligands

A appropriate targeting molecule on the cancer cell's surface and a corresponding high affinity ligand that binds to the targeted molecule precisely are needed for tumour targeting ligands. In general the goal of affinity targeting is for antibodies or their fragments, proteins, peptides, carbohydrates, or tiny receptor ligands coupled to CPPs to recognize cancer cell receptors. The interaction between receptors and CPP bounding ligands could be applied to targeting cancer tissue since cancer cells express a different repertoire of receptors to that of normal tissue. The creation of certain tumour homing peptides and cancer cell piercing peptides has been a significant breakthrough in this regards.(Ruoslahti et al., 2005)(Bauleth-Ramos & Sarmento, 2021)(Desale et al., 2021)(Maity et al., 2020). A number of tumour tissue homing peptides have been identified Some examples are iRGD (CRGDKGPDC) that incorporates an integrin-binding RGD motif for recognition of the tumour endothelium(Teesalu et al., 2013). the CPP YTA2 which is coupled to cytostatic methotrexate (MTX)and the cyclic peptide cCPGPEGAGC (PEGA) which becomes penetrant when bound to pVEC and functions as a breast tumour homing peptide in mice(Lindgren et al., 2006)(Myrberg et al., 2008).

2.3. Peptide drug conjugate (PDCS)

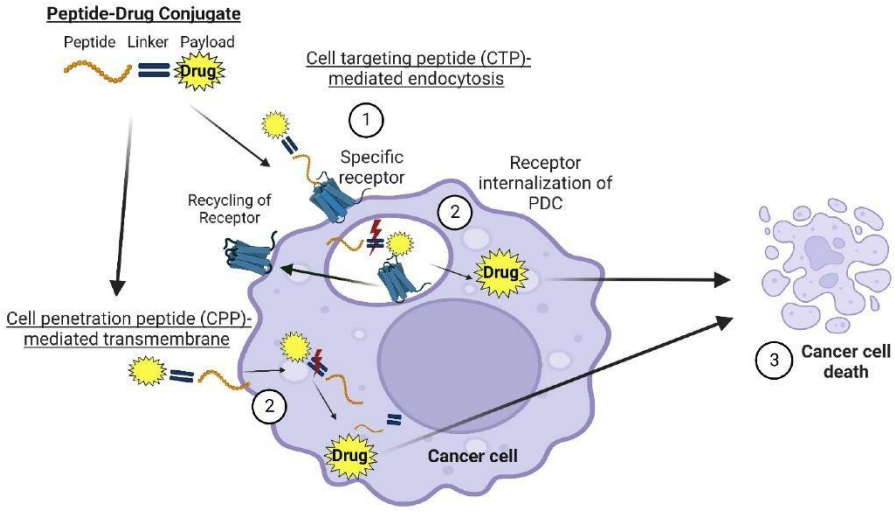


Figure2 peptide drug conjugate

Peptide drug conjugate (PDCs) are an emerging class of prodrug, formed trough the covalent attachment of a specific peptide sequence to a drug via a cleavable linker because of amino acid sequence may be selected to the both regulate the conjugates physiochemical characteristic and enable active targeting of a specific receptor on the tumour cell surface, the use of peptides enables the insertion of a high degree of functionally into PDCs. PDCs are biodegradable and shouldn't cause unwanted immunogenic reactions because they are made from amino acid and typically have short peptide length(Armstrong et al., 2025)(Fillingim, 2017). Self-assembling PDCs, in which the individual conjugates have the capacity to form nanostructures with physicochemical properties distinct from those of the components, are an emerging subset of PDCs that aims to combine the benefits of peptide-based prodrugs with those of a vehicular delivery approach(Su et al., 2015)(Ma et al., 2016). The drug retains its inherit bioactivity but also functions as building blocks in the construction of nano seized structure; additionally the peptide component acts like a conventional hydrophilic segment to stabilize the nanocarrier in water, but can also act as functional blocks for both formation and targeting purposes; manipulation at the molecular level also means that the numbers and locations of reactive site for drug conjugation can be precisely determined, leading the control over the structure of PCDs at the sequence level(Matsumura & Maeda, 1986).

2.4. peptide inhibitor of oncogenic pathways

2.4.1. KRAS

KRAS is the most frequently mutated and is responsible for about ~76% of RAS- mutated cancer, while NRAS accounts for 17% and HRAS for 7%. KRAS mutation are especially prevalent in cancer such as pancreatic adenocarcinoma (88%), lung adenocarcinoma (31%), and colorectal cancers (45-50%). In contrast, NEAS mutation are most associated with melanomas, acute myeloid leukemia, and thyroid carcinoma, whereas HRAS mutation are the most often seen in pheochromocytoma and paraganglioma, thymoma, and head and neck squamous cell carcinoma(Prior et al., 2020b). KRAS is the most frequently mutated genetic driver in lung cancer, according for approximately 26.1% of lung adenocarcinomas. In contrast, the prevalence of KRAS mutation in Asian population is lower, ranging from 10-15% surpassed by EGFR mutation(Dearden et al., 2013). Oncogenic KRAS mutation subtype are primarily classified based on the type of nucleotide substitution: transversion mutation (G→T or G→C), such as G12C, G12V, G12A, and G12R, and transition mutation (G→A), such as G12D, G12S, and G13D. over 80% of the KRAS Mutation occur at codon 12, Where the glycine amino acid is substituted, most commonly by cysteine (G12C), followed by valine (G12V) and aspartic acid (G12D)(Veluswamy et al., 2021). The distribution of these mutations subtype also varies significantly across different cancer types, reflecting the complex and heterogeneous nature of KRAS-driven cancers. KRAS glycine-to-cysteine(G12C) mutation accounts for approximately 13% of all lungs adenocarcinoma patients, while KRAS glycine-to-aspartic acid (G12D) mutation if found in 4% lung adenocarcinoma patients(Veluswamy et al., 2021).

2.4.2 EGFR

EGFR is a transmembrane receptor with tyrosine protein kinase activity encoded by the ErbB-1 gene, has three protein structural domain including and extracellular ligand binding structural domain, a hydrophobic transmembrane structural domain, and a cytoplasmic tyrosine kinase structural domain. The EGFR gene is located on chromosome 7, region 7p12, with a total length of 200kb, and is widely distributed in the cell membrane of various tissues of human body (Jiromaru et al., 2022). The EGFR inhibitor can be broadly categorized into two groups: monoclonal antibodies (mAbs) and small molecule tyrosine kinase inhibitors (TKLs). mAbs act extracellularly and avoid binding of EGFR ligands, whereas small molecules TKIs compete with kinase domain of ATP- binding receptor. The most commonly used EGFR inhibitors are mAbs. mAbs bind to the extracellular domain of EGFR, promoting its endocytosis, downregulation, and degradation(Xiong et al., 2025)

2.4.3. c-MET

Cellular-mesenchymal epithelial transition factor (c-Met), also named hepatocyte growth factor receptor (HGFR), is a special member of the RTK family. It consist of semaphoring (SEMA) domain, a plexin-semaphorin-integrin (PSI) domain, and four consecutive extracellular immunoglobin-plexin transcription factor (IPT1-4) domain, a single transmembrane helix (TM), and an intercellular KD. The c-Met can be activated by either its homologous ligands, hepatocyte growth factor (HGF), or its native subtype NK1. The combination of c-Met and HGF includes the dimerization of c-Met, leading to autophosphorylation of tyrosine residue in c-Met intracellular region and subsequent activation of downstream signaling pathways. Finally, the signal is transferred to the nuclear transcriptional machinery, regulating many essential cellular processes such as cell proliferation, motility, morphogenesis, and survival. C-Met inhibitors are commonly used to treat cancers by blocking the interaction between ligands and receptor, thereby preventing downstream signal activation through targeted inhibition of c-Met or HGF. However, due to the complexity of cancer, manipulating a single target may result in treatment failure(Jin et al., 2024). For example, type I c-Met kinase inhibitor are ATP-competitive. The affinity of inhibitors and c-Met becomes weaker owing to the pocket point mutation in the ATP binding pocket of c-Met, which then occur. Eventually, cancers treated with type I c-Met kinase inhibitor often became resistant. Drug combination for the treatment of cancers has proven to successful strategy, but the use of two or more drugs simultaneously reman challenge in clinical therapy because of unfamiliar technology, pharmacological properties, and the drug drug interaction (Hu et al., 2020). With the progress of related studies, research have gradually explained the multiple signaling pathways mediated by c-Met c-Met/EGFR, c-Met/ALK, c-Met/AXL, c-Met/HDACs, etc. In recent years researches have increasingly focused on developing novel drug-target drugs due to the advantage of drugs with multiple pharmacological activities over drug combination therapy(S.-W. Zhang et al., 2020).

2.5. Peptide immune modulators

2.5.1. PD1/PD-L1

The PD 1 complex is an inhibitory immunological check point that is crucial for immune control. The PD1 axis also responsible for cancer cell immune evasion, making it one of primary target of oncology immunotherapies. Immune checkpoint inhibitor therapy is primarily based on antibodies. This technique has enormous therapeutic. potential yet, it is significant disadvantage and can generate immune-related adverse effects. Thus, there is a strong needed for alternative, non-antibody-based therapies using small molecules, peptides, or peptidomimetics. In the present study, we designed, synthesized, and evaluated a set of Pd-1-

targeting peptides to PD-1 was investigated using SPR and ELISE. We also assessed their ability to complete with PD-L1 for binding to PD-1 and their inhibitory properties against the PD-1/PD-L1 complex at cellular level. The best result were obtained for the peptide PD-L1, named(L11), which displaced PD-L1 for binding to PD-1 in the competitive assay and inhibited the formulation of the PD-1. NMR investigations related the (L11) lacks a well-defined secondary structure yet, MD simulation showed that (L11) binds to PD-1 in the same location as PD-L1. After further structural optimization, the peptide inhibitor discovered in this study could be exploited as a potential therapeutic drug targeting the PD-1/PD-L1 axis(Bojko et al., 2024).

2.5.2. CLTA-4

CTLA-4 has a strong regulatory role in the early stages of T cell activation and is primarily expressed on activated CD8+ and CD4+ T cells, as well as constitutively on Treg cells (L. Chen & Flies, 2013)(Schnell et al., 2020). Blocking CTLA-4 can reverse and restore reduced T cells function, increase T cells proliferation and effector capacity, and markedly decrease tumor growth .The FDA has formally approved ipilimumab, a CTLA-4 monoclonal antibody, for the treatment of unresectable stages III/IV metastatic melanoma. irAEs from anti CTLA drugs are dose dependent and occur more frequently, limiting their therapeutic use(Zhou et al., 2022).We first obtained CTLA-4 affinity peptide LC4 by phage display peptide library screening in the study. LC4 peptide has been demonstrated of effectively block CTLA-4/B4 protein interactions and has good anti-tumour effect in vitro and in vivo. PLGLAG was used to connect LC4 with RGD, a tumour- targeting peptide sequence with high affinity for integrin αvβ3, and obtain a modified peptide LC4-PLG-RGD. LC4-PLG-RGD as a result has better anti-tumour effect, could reduce peripheral immune system activation, and reduce irAEs produced in CTLA-4 treatment.(Zhou et al., 2022)

2.6. Peptide vaccines and neoantigen-based approaches 4

Novel methods to triggering tumour-specific immune response in lung cancer involve peptide vaccines and neoantigen-based immunotherapies. Neoantigen are peptides that have outstanding tumor specificity and few adverse reactions that are produced from tumor-specific mutations shown on the molecules of major histocompatibility complex (MHC) and that T cells detect. Neoantigen peptide vaccines generated from the lung cancer translatome show impressive anticancer promise in vivo by triggering high immunogenicity and significant inhibiting tumor development in murine lung cancer models in preclinical researches(Lian et al., 2024). According to a recent studies, some neoantigens may improve vaccination efficacy by enhancing tumour specificity T- cell responses. For example, The Ddx21 mutation peptide simulated long-term anti tumor immune responses in lung cancer(Z. Zhang et al., 2025). Personalized neoantigen peptide vaccines have

been tested clinically inpatients with advanced non-small cell lung cancer (NSCLC), Where vaccination was safe, effective and produced neoantigen-specific CD8+ T cell responses. The objective tumor regression were noted in certain patients with EGFR-mutated tumors after receiving the vaccine(F. Li et al., 2021).

2.7. Stabilized peptide and peptidomimetics

Stabilized peptide and peptidomimetics are a significant and emerging class of lung cancer treatment because they may target protein-protein interaction (PPIs) that promote oncogenic signaling but are often inaccessible to conservational small molecule medicine. Example include interaction between the tumor suppressor P53 and its negative regulators MDM2/MDMX angiogenesis and invasion related integrin signaling and other intercellular complexes required for survival and proliferation. Chemical modification such as hydrocarbon cyclization and stapling have been show to significantly improve pharmacokinetic properties, protease resistance and structural stiffness(Walensky & Bird, 2014). Based on studies on stapled peptides and PPI inhibitor stapled peptide is the most effective ways to improve biological efficiency in cancer models and strengthen peptide therapies against degradation(Iyer, 2016). A major therapy for many solid cancers including NSCLC is reactivating the tumor suppressor p53 which is frequently inhibited by overexpression of MDM2 and MDMX rather than mutation. Stapled peptide based on the P53 N-terminal transactivation domain (such as SAH-P53 analogues) have demonstrated substantial disruption of MDM2-p53 connections and enhanced alpha-helicity when compared to their linear counterparts, indicating their potential as molecular inhibitors of oncogenic PPIs(Y. Li et al., 2024). Peptidomimetics based on cyclized or otherwise stabilized frameworks are attractive as next generation cancer therapeutics because they can reproduce the beta-sheet or alpha-helical patterns seen in oncogenic PPIs in addition to improving peptide stability(Laxio Arenas et al., 2019). To interact with αvβ3/αvβ5 integrins associated with tumor invasion and angiogenesis two crucial routes in development of medications for lung adenocarcinoma integrin targeted cyclic RGD peptidomimetics(Karnati & Anusha, 2025).

2.8. Diagnostic/imaging peptides (PET/SPECT/MRI/near-IR)

Positron Emission Tomography (PET) stands as the ideal modality for-invasive quantitative in vivo assessment of TROP-2 expression(X. Zhang et al., 2025). Among radionuclides, gallium-68 offer generator based production, a suitable 68-min half life and versatile chelation chemistry with NOTA or DOTA. In recent years, ^{68}Ga-based probes, including (^{68}Ga)Ga-NOTA-T4(Izumchenko et al., 2017) and ^{68}Ga-MT6349 have exemplified its translational potential for molecular imaging. These recent developments highlight the growing clinical interest in ^{68}Ga-labelled peptide tracers. Currently, TROP-2-targeted PET probes are designed based

on antibody agents; however, the large size of this macromolecules inherently limits their pharmacokinetic properties, manifesting as poor tissue penetration and prolonged systemic circulation(Y. Ahmed et al., 2021)(Omori et al., 2022). This limitation results in delayed tumor visualization and elevated background signaling, Which compromise imaging quality and hider the clinical application of such probes in heterogeneous tumors characteristic of NSCLC. In contrast, peptide-based probes emerges as a highly promising alternative. Their low molecular weight not only enhances tissue penetration efficiency but also enables rapid systemic clearance, in addition, their structural simplicity facilitates both straightforward chemicals modification and efficient radiolabelling(Fang et al., 2025)(Mankoff et al., 2017). The design of novel peptide-based PET probes on ligand-based computational design (LBCD) technology for TROP-2's natural ligand of IGF-1(Yu et al., 2025). A lead peptide sequence named RS15 was acquired. Following conjugation of RS15 with a NOTA chelator via a OEG_4 linker, efficient radiolabelling with gallium-^{68}(68Ga) produced [68Ga]Ga-NOTA-PEG_4-RS15(Yu et al., 2025).

3.Molecular targets and Mechanism of Action

3.1. Surface receptor targeting in lung tumors and microenvironment

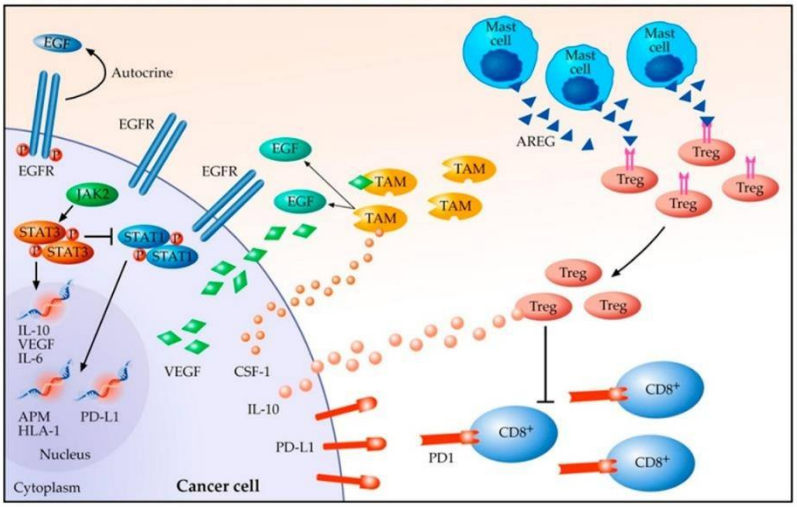


Figure 2

Surface receptor targeting in lung tumors and microenvironment

Surface receptor targeting has emerged as crucial elements of modern lung cancer treatments by using membrane proteins that are aberrantly expressed on tumor cells and cells in the tumor microenvironment (TEM). This makes it possible to specifically alter malignant signaling pathways and immune evasion. Receptors such as the epidermal growth factor receptor (EGFR), anaplastic lymphoma kinase (ALK), and ROS1-classical oncogenic driven in non-small cell lung cancer (NSCLC) whose improper activation enhances proliferation, invasion, and survival- have been successfully addressed by tyrosine kinase inhibitors. When compared to chemotherapy alone, patient outcomes have improved significantly(Hassanein et al., 2021)(Anu et al., 2025). Tumor-induced T-cell suppression has also been reversed by immunological related surface receptors in the TME, particularly the PD-1/ PD-L1 immune checkpoint axis. Immune receptors targeting is essential for overcoming immunosuppressive microenvironmental constrain as demonstrated by the increase cytotoxic T-lymphocyte activity and improved survival in advanced non-small cell lung cancer (NSCLC) when PD-1 or PD-L1 is blocked with antibodies such as pembrolizumab and nivolumab(Araghi et al., 2023). To improve innate and adaptive anti-tumor responses, TME components like regulatory T cells and tumor associated macrophages (TAMs) express receptors like scavenger receptors and C-type lectins in addition to direct tumor cell receptors. Targeting these receptor can repolarize TAMs from tumor- promoting (M2) to tumor-suppressing phenotypes(Sabir et al., 2021).Recently, receptor targeting has been used in tailored cell treatments and nanomedicine. To overcome TME barriers and diverse antigen expression, for instance, CAR-T cells that are designed to target tumor antigens like EGFR, Mucin-1, and mesothelin, as well as nanoparticles adorned with ligands for receptors like TAS1R3 in metastatic lung cancer, are available(Alenezi, 2025).

3.2. Intracellular pathway interception and synthetic lethality

Intracellular route interruption and synthetic lethality are two very promising precision oncology strategies for lung cancer, particularly in genetically defined subtypes such as non-small cell lung cancer (NSCLC). Lung cancer often involves changes to oncogenic drivers such as EGFR, KRAS, TP53, STK11, and ALK rearrangements. Important intracellular signaling cascades such as the RAS/RAF/MEK/ERK and P13K/AKT/mTOR pathways with small molecules inhibitors or targeted biologics tumor development can be stopped. These pathways control metabolism, DNA damage repair, apoptosis resistance, and proliferation. However, pathway redundancy and adaptive feedback loops often lead to acquired resistance. Synthetic lethality has emerged as a sensible technique to overcome this constrain. A genetic interaction known as “synthetic lethality” occur when two genes are altered simultaneously, although each genes alone does not result in cell death. One flaw in cancer treatment, like an oncogenic mutation, is unique to the tumor. Therapy

causes the second defect. This method preserves healthy tissue while eliminating malignant cells(Byrne, 2010).

3.3. overcoming efflux, hypoxia and stromal barriers

Treatment resistance in lung cancer is greatly influenced by three primary biological barriers stromal features of the tumor microenvironment (TME) drug efflux pumps and tumor hypoxia. The efficacy of immunotherapies, targeted treatment and chemotherapy is limited by these obstacles. ATP binding cassette(ABC) transporters, including ABCB1(p-glycoprotein/MDR1), ABCC1(MRP1), and ABCG2(BCRP), are the primary causes of efflux-mediated resistance. They actively remove chemotherapeutics agents from tumor cell through ATP-dependent mechanism, lowering intracellular drug accumulation and cytotoxicity. Efflux pump inhibitor that reduce ATP availability RNA interference that target transporter expression and delivery technique based on nanocarriers that go around transporter recognition that enhance intracellular drug retention(Seton-Rogers, 2016)(Seton-Rogers, 2016). Another important barrier in lung cancer is tumor hypoxia which is caused by abnormal vasculature and fast tumor growth. Hypoxia-inducible factor 1-alpha (HIF-1alpha), which transcriptionally activated genes related to angiogenesis (VEGF) glycolysis (GLUT1), epithelial-mesenchymal transition (EMT), and immune evasion is stabilize in hypoxia microenvironmental. In NSCLC, persistent HIF-1alpha signaling increases resistance to platinum-based chemotherapy and radiation HIF-1alpha inhibitors (like PX-478), hypoxia activated prodrug(like Tirapazamine), and agents that target tumor angiogenesis (like Bevacizumab) are al used to target hypoxia-activated prodrug work by producing cytotoxic radicals only in hypoxia tumor areas while sparking normoxia tissue through bio reductive activation at low oxygen tension. HIF-1alpha inhibition increases sensitivity to immunocheckpoint inhibitors and chemotherapy, decreases VEGF-mediated angiogenesis and decreases metabolic adaptation(Kirn & Thorne, 2009). The stromal barrier in lung cancer includes cancer-associated fibroblasts (CAFs), extracellular matrix(ECM) component like collagen and hyaluronic acid, tumor associated macrophages (TAMs), and dysfunctional vasculature. CAFs produce transforming growth factor-beta (TGF-beta), hepatocyte growth factor (HGF), and matrix metalloproteinases (MMPs), which alter the EMC and trigger oncogenic signaling pathway like P13/AKT and MAPK in cancer cells. Therapies targeting the stromal components includes TGF-beta inhibitor like galunisertib, focal adhesion kinase (FAK) inhibitors blocking tumor stroma interaction and enzymatic degradation of hyaluronic acid PEGylated hyaluronidase. Targeting the stroma mechanistically enhance drug delivery, raises vascular perfusion, lowers interstitial fluid pressure and modifies immunological infiltration by reducing TAMs and myeloid-derived suppressor cells.

Clinical studies have demonstrated that stromal-targeting drugs and immune checkpoint inhibitors such as nivolumab enhance antitumor immunity in advanced non-small cell lung cancer(Chang et al., 2017).

3.4. Immuno-oncology mechanism: T cell priming checkpoint interference, TAM reprogramming

In immune-oncology for lung cancer, effective antitumor immunity depends on antigen-presenting cells (APCs) the lymph nodes successfully priming naive T cells. Immunosuppressive signals in the tumor environment disrupt this mechanism, which causes effector CD8+ T cells to become activated and infiltrate the tumor microenvironment. Immunocheckpoint blocking (ICB), particularly targeting the PD-1/PD-L1 axis has transformed the therapeutic therapy of non-small cell lung cancer (NSCLC) by restoring effector function and revitalizing worn-out T cells. Though three are restriction due to resistance mechanism and immune exclusion phenotypes that impede T-cell priming and tumor infiltration, current molecular studies explain how PD-1/PD-L1 and CTLA-4 inhibition restores T-cell cytotoxicity(Wei et al., 2024)**.** The immunosuppressive role of TAMs in lung cancer microenvironment, where M2-polarized macrophages generate cytokines (such as TGF-beta and IL-10) and metabolic restrictions that lead to T-cell malfunction. To improve the efficacy of checkpoint inhibitors, they also outline novel strategies that aim to rewire TAMs to display pro-inflammatory M1 phenotypes or reduce M2 dominance(Veluswamy et al., 2021). To overcome intrinsic and acquired resistance to checkpoint blockade in non-small cell lung cancer, combination immunotherapeutic approaches that combine TAM modulation, immune metabolism targeting and dual checkpoint inhibition that interact with T-cell priming and effector responses are being researched(Shi et al., 2025).

4. Formulation and delivery strategies

4.1. Systemic delivery: PEGylation, albumin-binding, long-acting designs.

Systemic delivery optimization is still an important aspect of enhancing the therapeutic index of anticancer medications by extending their duration in the body, minimizing off-target toxicity, and boosting tumor accumulation. PEGylation is still being explored extensively because it increases hydrodynamic size and steric shielding, slowing enzymatic breakdown and immunological clearance and potentially extending systemic half-life. A review published in 2024 discussed how PEGylation of medicinal proteins and peptides has improved. It claims that stability has improved, immunogenicity has decreased and bioavailability has increased(C. Li et al., 2024). Albumin binding techniques employ endogenous transport channels to target tumors and maintain long term circulation. Endogenous albumin is an ideal carrier due to its long half-life and tendency to spontaneously collect tumors via the increased permeability and retention (EPR) effect. Drug

platforms linked to albumin have demonstrated enhanced pharmacokinetics and the possibility for better delivery to tumor tissue. A number of recent clinical and preclinical investigation have identified albumin-based nanocarriers as effective extended delivery system that improve medication stability and biodistribution while minimizing systemic toxicity generally associated with free medicines(Qu & Song, 2024). To maintain therapeutic concentration and lower dosing frequency, long-acting delivery system between 2022 and 2025 will increasingly incorporate prodrug design, injectable deposits, lipid-polymer hybrid nanoparticles and biodegradable polymers (like PLGA). In order to improve penetration recent studies highlight the significance of release- kinetic modelling, microenvironment responses cleavage mechanism (PH, redox, enzyme-triggered) and multistage nanocarriers that shrink or change after tumor accumulation. 2024 review addresses that the logical design of log-acting nanomedicines with programmable release profiles, emphasizing how surface functionalization, molecular weight distribution and polymer degradation rate regulate systemic pharmacodynamics(Duan et al., 2023).

4.2. Nanocarrier: liposomes, polymeric nanoparticles, micelles, dendrimers, hybrid biomimetic carriers

Nanocarrier have change the administration of cancer medications by increasing therapeutic efficacy, reducing systemic toxicity, and enabling targeted and controlled release of therapies.

Each of the many carrier types liposomes, polymeric nanoparticles, micelles, dendrimers, and hybrid or biomimetic systems offers special advantages and methods for formulation. By encasing hydrophilic and hydrophobic medication in their bilayer structure, phospholipid based vesicular system known as liposomes improve solubility, protect labile chemicals and use the increased permeability and retention (EPR) effect to promote tumor growth. Research is ongoing to modify surface chemistry and ligand conjugation for active targeting and stimuli responsive release. Clinically approved liposomal chemotherapeutics have longer circulation and fewer off-target toxicity than free medicines(Goshisht et al., 2025). Polymeric nanoparticles (PNPs) are multifunctional carrier that can be utilized for sustained drug release, surface enhancement for receptor mediated targeting and encapsulation of small molecules, peptide or nucleic acids. They composed of biodegradable polymers such as PLGA. PEGylated copolymers and stimuli-responsive materials. Particularly when tailored for PH or redox sensitive in the tumor microenvironment, PNPs have shown promise in regulated release, enhanced tumor retention, and decreased systemic cytotoxicity(Xiao et al., 2022). Micelles are created when amphiphilic block copolymers self-assemble in aqueous media. They have a hydrophobic core that makes poorly water-soluble chemotherapies more soluble They can also be designed for active targeting and stimulus-triggered breakdown, which allows for quick drug release in response to

tumor specific indicators like enzyme expression or an acidic PH(Zeinali et al., 2025). To improve intracellular absorption, overcome chemoresistance, and co-delivery medications and genetic material in malignancies such as ovarian carcinoma, modern design methodologies including tumor-targeting ligands and stimuli-responsive connections. Dendrimers are monodisperse, highly branched polymeric macromolecules with many terminal groups that provide precise surface modification and high drug loading capacity, allowing for multifunctionality and ligand conjugation(Zeinali et al., 2025). Drugs can more easily avoid the immune system, remain in the body longer, and target certain cells with hybrid biomimetic carriers, which combine biological membrane or component with synthetic nanomaterials. The structural advantages of both lipidic and polymeric components can be combined with peptide targeting moieties in lipid-polymer hybrid systems and other biological platforms. They are therefore more stable and less likely to trigger an immunological reaction, making them multipurpose vectors for medication delivery and diagnostics(Karati et al., 2024). These peptide-based nanocarrier approaches, which enable targeted drug administration, enhanced imaging contrast, controlled release, and therapeutic untegration, represent a potential development in lung cancer diagnostic and treatment. The problems of peptide stability in vivo, proteolytic degradation, and larger-scale manufacturing are the main focus of current study. Multimodal and stimuli-responsive nanoplatforms for individualized cancer therapy will be the focus of future developments(Omidian, Cubeddu, et al., 2025).

4.3. ligand decorative and multivalency (dual-targeting peptide shells)

Multivalency and ligand decorating have been identified as crucial formulation techniques to raise the therapeutic index and specificity of nanocarriers in the treatment of lung cancer. Specific binding to overexpressed receptors like integrins ($\alpha v \beta 3$), EGFR and folated receptors in non-small lung cancer (NSCLC) is made possible by surface modification of nanoparticles with target peptides like RGD (arginine-glycine-aspartic acid), TAT, GE11(EGFR-binding peptide). When compared to monovalent nanocarriers, multivalency the appearance of multiple similar or different ligands on a single nanocarrier can greatly boost avidity through cooperative binding interactions, leading to greater cellular uptake, receptor aggregation and endocytosis. Dual-targeted peptide coatings in which two ligands are covalently attached to liposomes, polymeric nanoparticles, or lipid polymeric hybrid nanocarriers have been shown in recent studies(Zhao et al., 2023). Similarly, mesoporous silica-liposome core-shell nanoparticles coated with both floated and EGFR improved cytotoxic drug delivery and receptor-mediated endocytosis in NSCLC(Rathnayake et al., 2023). In lung tumor models polymeric nanoparticles functionalized with targeted peptides including PLGA or chitosan, have demonstrated enhanced biodistribution and decreased systemic toxicity(X. Zhang et al., 2021).

4.4. Inhalable and intratracheal delivery (dry powder, nebulized solutions)

Because inhalation and intratracheal delivery strategies leverage the lung large surface areas, thin barrier and rich vasculature to maximum drug deposition at the tumor site while minimizing systemic toxicity they are being investigated as targeted formulation and delivery methods for lung cancer therapies. These strategies include direct pulmonary instillation, nebulized aerosols and dry powder inhalers (DPIs). Compared to intravenous and oral administration, pulmonary delivery allows for the development of high local drug concentration with minimal first-pass metabolism and systemic toxicity. This is particularly helpful for anticancer medications that are cytotoxic or poorly bioavailability(Peng et al., 2024). Due to its stability, mobility, ease of storage and lack of propellants, dry powder inhaler (DPI) formulations are a commonly used strategy for lung cancer treatment, researchers have recently created inhalable dry powders with polymeric nanoparticles encasing the kinase inhibitor crizotinib. These products have shown improved pulmonary deposition and release profiles with the potential to lower systemic toxicity, as well as controlled particle design adapted to aerosolization by devices like DPIs(Naureen et al., 2025). In order to produce "nano-in-micro" structured particles with ideal aerodynamic sizes (1-5um) for deep lung delivery, more improved DOI devices may employ particle engineering techniques like spray drying, spray freeze drying, and freeze drying. This can enhance lung retention and the bioavailability of chemotherapeutic medications like paclitaxel(Cao et al., 2024). Nebulizer-generated liquid aerosols can serve as an alternated inhalation route particularly for suspensions of nanocarriers like liposomes and solid lipid nanoparticles. It has been demonstrated that nanoparticle formulations may preserve structural integrity during aerosolization and nebulizers can generate respirable aerosolized droplets that can distributed nanoparticle-based medication evenly throughout the bronchial tree(W. Wang et al., 2024). The clinical unity of inhalation approached for lung cancer chemotherapy is supported by evidence from inhaled liposomal chemotherapeutics, such as aerosolized liposomal 9-nitro camptothecin, which demonstrate that nebulized delivery can provide significantly higher drug concentration in the lung with tolerable systemic exposure(Safdar et al., 2025). In preclinical research, intratracheal administration which is more invasive if frequently used to guarantee homogenous tumor distribution and enable direct delivery of drug formulation to lower the respiratory tract. Intratracheal injection has been demonstrated to ensure region- specific delivery and prolonged release of anticancer medicined harm to normal tissue in preclinical trials employing dry powder nano and microparticle formulations(Mangal et al.,

2017). Formulation elements like particle size control (1-5um aerodynamic diameter), excipients, carrier design (lipid polymer or hybrid), aerosolization stability and resistance to rapid mucociliary and phagocytic clearance are crucial for optimizing deposition and therapeutic effects in the lung in al of these pulmonary delivery techniques(Peng et al., 2024).

4.5. Localized delivery (hydrogels, implants)

Localized delivery strategies, such as hydrogels and implantable sites, are becoming important formulation and delivery approaches in lung cancer therapy because they enable direct drug deposition at the tumor site, sustain release, reduce systemic toxicity and enhanced therapeutic efficacy when compare to traditional systemic administration. In order to minimize off target effects and deliver a high local drug concentration within the tumor microenvironment, hydrogels biocompatible, water-rich networks of polymers can encapsulate biologics, chemotherapeutic agents or nanocarriers and form localized depots in place after injection or implantation. Injectable and in place forming hydrogels often undergo a sol-gel phase shift from a minimally invasive liquid to semi-solid matrix that remain at the tumor site for prolonged drug release when physiological factors (such PH or temperature) trigger this transformation(D.-Y. Fan et al., 2019). For usage in lung cancer-specific applications, porous hydrogel scaffolds composed of biopolymers such as sodium carboxymethyl cellulose (SCMC) and gelatine have been created. Doxorubicin can be locally administered straight into lung tumor locations thanks to these structures. The high drug encapsulation efficiency, prolonged release patterns and effective suppression of lung cancer cell growth in vitro of these hydrogel depots allow for the maintenance of therapeutic drug levels at tumor site while reducing systemic exposure(Chittasupho et al., 2021).Additionally, in preclinical studies, injectable self-healing hydrogels made from natural polymers like carboxymethyl chitosan and oxidized sodium alginate have demonstrated low systemic toxicity, prolonged intratumoral release and improved anti-tumor efficacy in the controlled local delivery of targeted local delivery of targeted therapeutic like anlotinib in non-small cell lung cancer(NSCLC)(Lan et al., 2025). These smart hydrogels can also be designed with mechanical strength and adhesive properties that encourage retention in the tumor microenvironment and extend medication exposure to enhance treatment outcomes.(Lan et al., 2025). In addition to chemotherapeutics drugs, thermosensitive and stimuli-responsive hydrogels which transform into a gel in reaction to body temperature or tumor specific stimuli have been studied for the treatment of lung cancer. For instance, thermosensitive hydrogels encasing targeted medicines like as erlotinib shows improved drug loading, longer retention around the tumor, higher local concentrations, and less systemic toxicity as compared to oral intervention delivery. The responsiveness of injectable hydrogels to physiological cues like as temperature, PH, or enzymatic condition allows for controlled release kinetics that

are particular to the tumor microenvironment. This increases effectiveness while reducing unintentional harm to healthy tissues(Sun et al., 2025). Implantable localized depots, which are made of polymeric matrices or in situ-forming implants inserted at or near lung tumors to distribute encapsulated medication in a sustain way, are another type of localized delivery system in addition to hydrogels. Once administered these implants may produce a solid or semi-solid depot, further limiting systemic exposure. The potential of injectable and implantable hydrogel platforms for localized chemotherapy to increase the therapeutic index by preserving high dose concentration in the tumor microenvironment while reducing systemic toxicity has been highlighted by studies on these systems(Pouso et al., 2025).

4.6. Co-delivery of small molecules, siRNA/miRNA and CRISPR cargo with peptides

In lung cancer treatment, the co-delivery of nucleic acid therapeutics (siRNA, miRNA, and more recent CRISPR gene editing payloads) and small molecule chemotherapeutic agents is a complicated formulation approaches that aims to simultaneously target multiple oncogenic pathways and resist drug resistance, treatment responses heterogenicity and poor cellular uptake of single agents therapeutics. Multifunctional nanocarrier such as lipid nanoparticles (LNPs) polymeric nanoparticles peptide modified carrier, and hybrid lipid-polymer particles have been designed to simultaneously encapsulated hydrophobic chemotherapeutic agents and hydrophilic nucleic acids in a single carrier system shields sensitive genetic payloads from nuclease degradation while achieving coordinated delivery and release in cancer cells(C. Zhang et al., 2020). Targeted polymer-peptide conjugates, including PLGA nanoparticles modified with the LHRH ligand, have also been designed to delivery curcumin with BCL2-targeting siRNA in order to address both pump and non-pump causes of chemoresistance, better cellular internalization, prolonged release, downregulation of genes linked to resistance and preferential lung accumulation are the outcomes of this(Lahan et al., 2024). The creation of inhalation-compatible nanocomplexes, such as fullerene-based siRNA carrier that carry PD-L1 siRNA by aerosol to lung demonstrates the viability of pulmonary co-delivery of genetic cargo in lung cancer. This prolongs pulmonary retention, stops siRNA from degrading, and successfully stops the spread of metastatic cancers(S. Liu et al., 2023). Although CRISPR/Cas gene editing systems have mostly been used in oncology research due to delivery issues, nanotechnology-enable delivery platforms are emerging that encapsulate cas9 protein or mRNA and sgRNA for precise gene editing to disrupt oncogenes or modify the tumor microenvironment. Preclinical evidence suggests that lipid and polymeric nanocarriers, which are intended to enhance intracellular slavery, endosomal escape and targeting may be used for lung gene editing(Rauf et al., 2025). Co-delivery may allow for simultaneously gene disrupt or correct and cytotoxic therapy, addressing both downstream proliferative signaling and the genetic drivers of lung cancer. This makes the integration of

CRISPR delivery with small molecule drugs, although still in its early stages, conceptually appearing for precision oncology. However, these co-delivery system use ligand/peptide targeting, endosomal escape strategies, stimuli-responsive release and tailored physiochemical design to overcome biological barriers and achieve synergistic therapeutic outcomes, even though issues like differential cargo stability controlled sequential release, immune activation and scalable manufacturing still need to e optimized for clinical translation(Deng et al., 2025).

5. Pharmacokinetic, biodistribution, and stability

5.1. Proteolysis resistance: D-amino acids, crystallization, stapling, N-methylation

Proteolytic instability significantly limits the pharmacokinetic performance of peptide-based therapeutic and peptide functionalize nanocarrier in lung cancer because endogenous protease in plasma, tumor tissue, and bronchoalveolar fluid rapidly breakdown linear L-amino acid peptide, resulting in short circulation half -life, rapid renal clearance and suboptimal tumor accumulation. To get over this limitation, several chemical and structural engineering methods have been extensively researched between 2019-2025. These methods include N-methylation, cyclization, hydrocarbon stapling and D-amino acid replacement. All of these methods raises enzymatic resistance and enhance pharmacokinetic and biodistribution characteristics. Recent studies on protease-resistance peptide treatments state that, when appropriately personalized the inclusion of D-amino acids or retro- enation peptide design significantly reduces protease recognisatin due to stereochemical inversion, extending serum stability and maintaining biological activity(Díaz-Perlas & Oller-Salvia, 2023). By limiting peptide conformations, reducing backbone flexible and protecting cleavage-prone regions, cyclization strategies such as head-to-tail macrocyclization and side-chain-n bridging enhance metabolic stability and in vivo exposure in comparison to linear equivalents(Soc et al., 2021). Similarly, because hydrocarbon stapling stabilizes α-helical structure , enhances membrane permeability, boosts resistance to enzymatic degradation, and extends systemic half-life, stapled peptides are promising candidates for targeting intracellular oncogenic protein-protein interactions in lung cancer(Moiola et al., 2019).

5.2. Tumor accumulation, EPR vs active targeting

Active targeting through ligand-receptor interactions and passive targeting through the increased permeability and retention (EPR) effect are two primary targeting concepts that support medication delivery using nanoparticles for the treatment of cancer. By allowing nanoparticles within an ideal size range (typically 10-200) to extravasate from blood vessels and accumulate especially in the tumor interstitial, the EPR effect

reduces systemic exposure and increases local drug concentration relative to free chemotherapeutics. It is caused by the aberrant and highly permeable vasculature of tumor tissue combined with inadequate lymphatic clearance. However, because of the substantial variability in human tumor vasculature, high interstitial fluid pressure, and strong extracellular matrix that prevent deep penetration and uniform distribution within solid tumors like lung carcinoma, its therapeutic performance has been variable. Despite significant outcomes in pre-clinical studies, this frequently leads to poor total tumor formation. This phenomenon continues to be a fundamental aspect of nanomedicine, helping to improve the bioavailability of encapsulated medication, increase their residency in tumor tissue, and extend circulation(Khan et al., 2026). Using specific ligands, such as peptides, antibodies, or small molecules, to modify the surface of nanoparticles that bind to overexpressed receptors on cancer cell or components of the tumor microenvironmental is known as active targeting. Although active targeting does not remove the need for initial extravasation through the tumor vasculature, it enhances selectivity and internalization beyond what passive EPR-driven accumulation alone can achieve by encouraging receptor-mediated endocytosis. Ligand-directed nanoparticles have demonstrated enhanced biodistribution profiles in preclinical cancer models, including better therapeutic indices, reduced off-target distribution and enhanced localization at tumor locations(Nguyen et al., 2021). Both passive and active targeting must be carefully planned from a pharmacokinetic perspective in order to maximize the stability and half-life of systemic circulation; surface engineering such as PEGylation, reduces interacting and reticuloendothelial system clearance, increasing the risk of tumor exposure and extending circulation. Active targeting adds another layer that can alter the kinetics of cellular distribution by promoting receptor-specific binding and retention, which often results in improved pharmacodynamic responses and greater intracellular drug delivery. However, significant barriers to clinical translation remain, including variables receptor expression across tumor subpopulation ligand stability in circulation, and potential immunogenicity of treated moieties(Vagena et al., 2025).

5.3. Lung deposition physics for inhaled peptides

Effectivity lung deposition of inhaled peptides for therapeutic purposes, including lung cancer, is mostly determined by the complex airway structure of the respiratory tract and the aerodynamic properties of particles. The size, shape, density, and hygroscopicity of inhaled particles define their aerodynamic diameter, which in turn determines the deposition site. While bigger particles tend to settle in the upper airways and smaller particles (less than 1um) are more likely to be expelled aerosols in the 1-3 um range are more likely to penetrate the deep lungs (alveolar area.) The effectiveness of treatment and local biodistribution are directly impacted by this. According to research, it is crucial to target peripheral lung areas by optimizing the mass

median aerodynamic diameter (MMAD) and fine particle fraction (FPF) when administering therapeutic peptides to tumors, aerodynamic properties also affect pharmacokinetics because the deposition site affects the peptide payload's absorption rates, residence time, mucociliary clearance am systemic uptake(R. Ahmed et al., 2025). Advanced inhalation technologies like spray-drying, dry powder inhalers (DPIs) and formulations assisted by nanocarriers have been shown to improve the stability and aerosol performance of peptides, improving pharmacokinetic profiles and localized biodistribution in lung tissues by preserving structural integrity and achieving controlled and repeatable lung deposition(Omidian, Nokhodchi, et al., 2025).

5.4. Bioanalysis methods and imaging readouts

When evaluating the pharmacokinetic biodistribution and stability of therapeutic drugs employed in the treatment of lung cancer biodistribution and stability of therapeutic drugs employed in the treatment of lung cancer bioanalytical technique and imaging readouts are essential. These methods allows scientists to examine the behaviour of anticancer medications, peptides, antibodies and nanocarrier systems in biological systems both statically and qualitatively. Drug absorption, distribution, metabolism and elimination are all measured in pharmacokinetic investigations. These factor are crucial for figuring out the best dosage schedules and therapeutic efficacy. Anticancer medications and their metabolites are often measured in biological matrices such as plasma, lung tissue and tumor samples, using sophisticated analytical methods including liquid chromatography combined with tandem mass spectroscopy (LC-MS/MS)(Werth et al., 2024). Molecular imaging techniques provides additional information to bioanalytical approaches by providing non-invasive viewing of medication distribution and tumor targeting in living animals. Positron emission tomography (PET) is one of most sensitivity imaging methods utilizing in lung cancer research. PET imaging uses radiolabeled tracers such as fluorine-18 or zinconium-89 to tracers accumulation in lung tumor tissues and allow for the quantitative assessment of metabolic activity, medication absorption, and receptor binding. Because it enables researchers and physicians to track therapy response and refine targeting treatments for non-small-cell lung cancer (NSCLC), PET imaging has grown in importance as tool in precision oncology(Q. Wang et al., 2022).

6. Safety, immunogenicity and toxicology

6.1. Membrane toxicity

In the development of lung cancer treatments, membrane toxicity is a significantly safety concern, especially for peptide therapies, immunotherapeutic agents and drug delivery system based on nanocarriers. Lipids bilayers, membrane protein and glycoproteins make up cellular membranes, which control signaling transport,

and cellular homeostasis. The structure, permeability and signaling pathways of these biological membrane may be changed by nanoparticles or anticancer medications, which could result in cytotoxicity or immune activation. The way that nanoparticles interact with biological membranes is largely determined by their physicochemical properties, including size, surface charge, composition and surface coating(Rashid et al., 2025). Therapeutic nanoparticles may cause immunogenic reactions that affect safety and treatment result in addition to causing direct membrane damage. The term immunogenicity describes a therapeutic agents or drug delivery systems capacity to activity the immune system, which may result in inflammation the release of cytokines, or quick removal of the therapeutic agents from circulation. The release of cytokines or quick removal of therapeutic agents from circulation. The reticuloendothelial system (RES) is activated in lung cancer treatment when immune cell like macrophages and dendritic cells identify nanocarrier as foreign particles. This immune activation may increase systemic toxicity and decrease drug accumulation in tumor tissues. Biomimetic techniques like cell membrane coated nanoparticles have been developed to address this problem(Y. Zeng et al., 2022).

6.2. Hypersensitivity and immunogenic mitigation

When it comes to peptide-based vaccines and immunotherapies that seek to stimulate the host immune system against tumor associated antigens, hypersensitivity and immunogenicity are major safety and toxicological are major safety and toxicology concerns in cancer treatment. By stimulating cytotoxic T lymphocytes and antigen-presenting cell, these therapies destroy specific tumor cell, but they may also have unanticipated side effects such hypersensitivity responses cytokines release, and the production of anti-drug antibodies. Immunological variability delivery mechanism, peptide structure and epitope makeup. Clinical and preclinical studies have shown that peptide-based cancer vaccines often have favourable safety profiles with few major side effects while yet generating substantial T-cell responses, highlighting their therapeutic promises(Esmaeili, 2025). To lower these hazards, a number of innovative methods that focus on both molecular design and delivery optimization have been created. Structural modifications such as peptide cyclization, PEGylation, and the insertion of non-natural amino acids can reduce immunogenic epitopes and increase stability. Rational epitopes selection and tailored neoantigen-based vaccination reduce the likelihood of autoimmune responses while increasing specificity. Moreover, cutting-edge delivery techniques like liposomes and nanoparticles are crucial for reducing systemic immune exposure, enabling regulated release, and preventing premature peptide breakdown. Additionally, it has been shown that combination approaches, including the use of immune checkpoint inhibitors and adjuvants, increase treatment effectiveness while more precisely modifying(Esmaeili, 2025).

7. Clinical translation and regulatory landscape

7.1. peptide therapeutics and diagnostics in lung cancer trials

Peptide therapeutics and diagnostics gained significant attention in lung cancer clinical trials because to their high specificity, ability to target tumor-associated receptors and typicaly superior safety profiles that conventional chemotherapy. Recant studies have shown that anticancer peptides can precisely interact with key oncogenic pathways such as EGFR, Alk, BRAF, and TP53, inhibiting tumor development and spread while minimizing damage to healthy tissues(Karankar et al., 2025). Peptide based vaccinations and immunotherapies can stimulate strong antigen-specific T-cell responses in patients with non-small cell lung cancer (NSCLC), according to phase I studies. In early-stage studies, objective tumor regression is still restricted(Suzuki et al., 2013). Additionally, peptide-drug conjugates and synthetic thymic peptides have shown better clinical results when used in conjugation with chemotherapy, including higher response rates advantages for survival and a reduction in chemotherapy related toxicities(F.-L. Zeng et al., 2019). Peptide-based technologies, such high-throughout peptide microarrays, which have demonstrated excellent sensitivity and specificity in detecting lung cancer biomarkers and tracking disease progression assist precision oncology and early detection(S. Yang et al., 2025).

7.2. Biomarkers strategies and patient

Biomarker techniques have become an important part of lung cancer research and clinical care because they enable early identification, patient classification, and personalized therapy selection. Lung cancer biomarkers include molecular (DNA, RNA mutations like EGFR, KRAS, ALK), protein-based, epigenetic, and circulating biomarkers like circulating tumor for diagnosis, prognosis, and treatment response monitoring(Qi et al., 2026). These biomarkers are increasingly being utilized to stratify patients into subgroups for targeted therapy and immunotherapy in non- small cell lung cancer, where biomarkers-driven treatment pathways are well-established. This improves clinical decision making and results (Patel et al., 2015). The clinical translation of these biomarkers still requires a number of processes, including discovery, analytical validation, clinical validation, and clinical value demonstration. Even while many biomarkers shown promise, only a few number of them effectively make it to normal clinical practice due to problem such tumor heterogeneity, poor biomarker abundance in early illness, lack of standardization and the need for substantial validation studies(Qi et al., 2026).

8. Comparative Effectiveness and positioning

8.1. Peptides vs antibodies, small molecules, and ADCs

Peptides, antibodies, small molecules, and antibody-drug conjugates (ADCs) are distinct but complementary therapeutic methods with their own advantages, disadvantages, and clinical use in lung cancer. Small drugs like EGFR and ALK tyrosine kinase inhibitors are commonly employed in the treatment of non-small cell lung cancer (NSCLC) because of their ability to infiltrate cells and disrupt intracellular carcinogenic pathways. However, they often have off target toxicity and rapidly become resistance(Patel et al., 2015). The advantage of mAbs is specificity and have greatly advanced immunotherapy while size limitations restrict penetration of the tumor. ADCs deliver a potent cytotoxic payload via the specificity of the antibody they can induce tumor killing and shown potential responses in biomarker-selected lung cancer, although also carry risks with toxicity and design complexity(Passaro et al., 2023). Therapeutic peptides represent a potential approach for same reasons due t their relatively small size, the high selectivity and improved tumor penetration. They also target oncogenic pathways including EGFR, BRAF and MET and have potential to minimize the systemic toxicity(Karankar et al., 2025).

8.2. Combination regimens (with TKIs, chemotherapy, radiotherapy, ICI)

Combination regimens integrating tyrosine kinase inhibitors (TKIs) chemotherapy, radiotherapy, and immune checkpoint inhibitors (ICIs) have become a cornerstone in the clinical management of lung cancer, driven by their ability to enhance therapeutic efficacy and overcome resistance mechanisms. In mutation-drive non-small cell lung cancer(NSXLC), TKIs are often used as first-line therapy; however, resistance inevitably develops, promoting the use of combination approaches. Since the immune checkpoint drug boost anticancer immune responses while chemotherapy lowers tumor burden, the combination off immunotherapy and chemotherapy has shown notable survival benefits in extensive-stage small cell lung cancer (SCLC)(Georgakopoulos et al., 2024). By encouraging tumor antigen release and boosting immune activation, radiotherapy also play critical synergistic role when combined with ICIs. This phenomenon improves response rates and is now included in standard treatment protocols like chemoradiotherapy followed by immunotherapy(Burr et al., 2024). Although issue like increasing toxicity, ideal sequencing and patient selection continue to be major obstacles, these combination regimens ae becoming more and more validated in clinical translation through adaptive and biomarker-driven clinical trials. From a regulatory standpoint, organization like the FDA and EMA support combination strategies through frameworks that prioritize co-development, companion diagnostics and integration of real-world evidence. However, approval pathways are still complicated because it is necessary to prove safety and synergistic efficacy across multiple agents. Although additional optimization and

regulatory harmonization are necessary for wider clinical implementation, combination regimens involving TKIs, chemotherapy, radiotherapy, and ICIs represent a paradigm shift in lung cancer treatment, offering improved survival outcomes and serving as the cornerstone of current precision oncology(Xie et al., 2023).

9. Challenges, knowledge Gaps and future directions

9.1. Target heterogenicity and resistance evolution

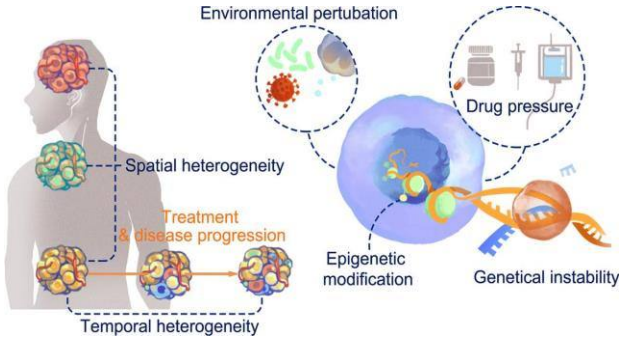


Figure 4

Targeting heterogenicity and resistance evolution

Fundamental obstacles to successful lung cancer treatment are tumor heterogeneity and resistance evolution, which result from intricate genetic, epigenetic and microenvironmental diversity both within and between tumors. Multiple subclonal populations with different molecular profiles coexist as a result of intratumoral heterogeneity, allowing for a Darwinian selection process under therapeutic pressure where resistance clones gradually proliferate(Zito Marino et al., 2019)(Cheng & Chen, 2014) By encouraging immune evasion, metabolic adaptation and protective niches that restrict drug penetration and efficacy, the tumor microenvironment (TME) plays a crucial role in the evolution of resistance(L. Liu et al., 2026). The lack of predictive biomarkers for therapy selection, especially in small-cell lung cancer (SCLC), the toxicity and complexity of combination treatment strategies and the incapacity of traditional biopsy techniques to fully capture spatial and temporal heterogenicity are some of the problems that persist despite significant advancements(Sen et al., 2024).

9.2. personalized peptide vaccines and real-time neoantigen selection

A novel approach to lung cancer immunotherapy is the use of tumor-specific mutations to generate highly tailored anti-tumor immune responses individualized peptide vaccines, and real-time neoantigen selection. Neoantigens, which are generated from somatic mutations unique to cancer cells, provide more specificity

than traditional tumor-associated antigens, enabling the activation of both CD4 and CD8T cells while lowering off target damage(X. Chen et al., 2020). Developments in immunopeptidomes, whole exome sequence, and next generation sequencing (NGS) have made it possible to identify and rank patient-specific neoantigens, which has made it easier to create synthetic long peptide vaccines that are customized to each tumor's unique mutational profile. Despite these promising developments, a number of barriers prevent their clinical use in lung cancer. Neoantigen selection is complicated by tumor heterogenicity and clonal evolution because sub clonal mutations may not be expressed consistently throughout all tumor sites, resulting in insufficient immune targeting. Furthermore, the accuracy of existing computational algorithms for predicting neoantigen immunogenicity is low, frequency failing to identify peptides that an elicit strong and long-lasting T-cell responses. The application of real-time neoantigen selection is further restricted by logistical and technological challenges, such as the time-consuming processes of sequencing data analysis, and customized vaccine production, which may not be compatible with the rapid development of advanced lung cancer. Additionally, immune checkpoint signaling, myeloid-derived suppressor cells, and regulatory T cells comprise the immunosuppressive tumor responses and encourage immune escape(Prabhu et al., 2026).

Conclusion

The limitations of conventional therapies like chromatography, radiation and surgery which often prove futile in advanced stages are addressed by peptide-based approaches for lung cancer detection and therapy. According to the report, peptides are highly versatile, biocompatible and particular molecules that are useful in both therapeutic and diagnostic contexts. They can precisely target overexpressed receptors including as c-MET, integrins, KRAS, and EGFR. This inhibits the growth of malignancies and oncogenic signaling pathways. Peptide inhibitors block key processes that aid in the growth and resistance of cancers, whereas peptide-drug conjugates (PDCs) allow cytotoxic medications to specifically target cancer cells. There are other use for peptides than direct targeting. Additionally they aid in immune system modulators. To lessen immune-related adverse effect, peptide inhibitor such as PD-1/PD-L1 and CTLA-4 can be employed in place of antibody-based checkpoint treatments. Neoantigen-based tactics and peptide vaccines enhance tumor-specific

immune responses, showing promise in NSCLC clinical studies. By improving pharmacokinetics, protease resistance, and structural stability, stabilized peptides and peptidomimetics successfully address a major obstacle in peptide treatments. Additionally, malignancies can be detected earlier and more precisely with diagnostic peptides for PET, SPECT, MRI and near-IR imaging that with bulking antibody tracers. Peptide-based approaches provide a multifaceted approach to lung cancer treatment by combining immune regulation tailored therapy, sophisticated diagnostics and intelligent delivery methods. Research into computational design, stable frameworks and combination medicines is increasing the likelihood of clinical success, even through issue like stability and heterogenicity still need to be addressed. Ultimately by improving the efficacy customized and safety of lung cancer therapies, these approaches may revolutionize the field.

Future prospect-

Given the rapid advancement in molecular biology, nanotechnology and precision medicine, peptide-based approaches to lung cancer diagnosis and treatment hold great promise for the future. The emergence of tailored peptide-based treatment, particularly neoantigen-derived vaccination that are tailored to each individual's tumor mutation profile using next-generation sequencing and bioinformatics techniques should will reduce toxicity to other cells while increasing tumor specificity and strengthening the immune response. Simultaneously, developments in peptide engineering-such as the synthesis of stapled peptides peptidomimetics, and the incorporation of artificial amino acid are improving the stability, bioavailability and therapeutic efficacy of peptide medications. This means they can now target intercellular protein-protein interactions they were previously impossible to drug Furthermore, it is expected that peptides would significantly enhance targeted drug delivery and controlled release inside the tumor microenvironment when

included into intelligent and stimuli-responsive delivery system, such as PH-sensitive, hypoxia-responsive and enzyme-activated nanocarriers. Another crucial areas of study is the combination of peptide therapies with other treatments such as immune checkpoint inhibitors, tyrosine kinase inhibitors, chemotherapy and radiation. This is due to its synergistic actions, which can aid in overcoming medication resistance and tumor heterogenicity. It is also anticipated that peptide-based immunotherapies would increase, particularly those that target immunological checkpoint like PD-1/PD-L1 and CTLA-4. They will provide alternatives to monoclonal antibodies that are safer and more efficient. Peptide-based imaging agents for PET, SPECT, MRI and near-infrared imaging are also anticipated to alter easily disease diagnosis and therapy monitoring due to their improved tumor penetration and quicker clearance. Targeting the lung directly while reducing the risk of systemic toxicity is made feasible and localized delivery technologies, such as injectable hydrogels, nebulized formulation, and dry powder inhalers. Furthermore, it is anticipated that novel approaches to overcome obstacles in the tumor microenvironment, including as hypoxia, stromal resistance and immune suspension, would improve treatment results. Peptides may be delivered concurrently with genetic materials like s RNA, miRNA, and CRISPR/Cas systems, opening up new possibilities for gene-based and multi-targeted treatments.

References-

Ahmed, R., Tewes, F., Aucamp, M., & Dube, A. (2025). Formulation and clinical translation of inhalable nanomedicines for the treatment and prevention of pulmonary infectious diseases. *Drug Delivery and Translational Research*, *15*(9), 2967–2993. https://doi.org/10.1007/s13346-025-01861-5

Ahmed, Y., Berenguer-Pina, J. J., & Mahgoub, T. (2021). The Rise of the TROP2-Targeting Agents in NSCLC: New Options on the Horizon. *Oncology*, *99*(10), 673–680. https://doi.org/10.1159/000517438

Alenezi, S. K. (2025). CAR T cells in lung cancer: Targeting tumor-associated antigens to revolutionize immunotherapy. *Pathology, Research and Practice*, *269*, 155947. https://doi.org/10.1016/j.prp.2025.155947

Alghisi, G. C., & Rüegg, C. (2006). Vascular integrins in tumor angiogenesis: mediators and therapeutic targets. *Endothelium : Journal of Endothelial Cell Research*, *13*(2), 113–135. https://doi.org/10.1080/10623320600698037

Anu, D., Rennis, K. D., Soumya, V. V., Jisna, B., Hussan, K. P. S., Chandrasekharan, L., & Babu, T. D. (2025). Targeting the Tumor Microenvironment in Lung Cancer with a Focus on Immune Evasion, Hypoxia, and Treatment Resistance. *Journal of Advanced Health Research & Clinical Medicine*, *2*(1).

https://journals.lww.com/hrcm/fulltext/2025/01000/targeting_the_tumor_microenvironment_in_lung.2.aspx

Araghi, M., Mannani, R., Heidarnejad maleki, A., Hamidi, A., Rostami, S., Safa, S. H., Faramarzi, F., Khorasani, S., Alimohammadi, M., Tahmasebi, S., & Akhavan-Sigari, R. (2023). Recent advances in non-small cell lung cancer targeted therapy; an update review. *Cancer Cell International*, *23*(1), 1–26. https://doi.org/10.1186/s12935-023-02990-y

Armstrong, A., Coburn, F., Nsereko, Y., & Al Musaimi, O. (2025). Peptide-Drug Conjugates: A New Hope for Cancer. *Journal of Peptide Science : An Official Publication of the European Peptide Society*, *31*(8), e70040. https://doi.org/10.1002/psc.70040

Barker, A. J., Gibson, K. H., Grundy, W., Godfrey, A. A., Barlow, J. J., Healy, M. P., Woodburn, J. R., Ashton, S. E., Curry, B. J., Scarlett, L., Henthorn, L., & Richards, L. (2001). Studies leading to the identification of ZD1839 (Iressa^{TM}): An orally active, selective epidermal growth factor receptor tyrosine kinase inhibitor targeted to the treatment of cancer. *Bioorganic and Medicinal Chemistry Letters*, *11*(14), 1911–1914. https://doi.org/10.1016/S0960-894X(01)00344-4

Bauleth-Ramos, T., & Sarmento, B. (2021). In Vitro Assays for Nanoparticle—Cancer Cell Interaction Studies. In *Advances in Experimental Medicine and Biology* (Vol. 1295). https://doi.org/10.1007/978-3-030-58174-9_10

Bessman, N. J., Freed, D. M., & Lemmon, M. A. (2014). Putting together structures of epidermal growth factor receptors. *Current Opinion in Structural Biology*, *29*, 95–101. https://doi.org/10.1016/j.sbi.2014.10.002

Bojko, M., Węgrzyn, K., Sikorska, E., Ciura, P., Battin, C., Steinberger, P., Magiera-Mularz, K., Dubin, G., Kulesza, A., Sieradzan, A. K., Spodzieja, M., & Rodziewicz-Motowidło, S. (2024). Peptide-based inhibitors targeting the PD-1/PD-L1 axis: potential immunotherapeutics for cancer. *Translational Oncology*, *42*, 101892. https://doi.org/10.1016/j.tranon.2024.101892

Bray, F., Laversanne, M., Sung, H., Ferlay, J., Siegel, R. L., Soerjomataram, I., & Jemal, A. (2024). Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries. *CA: A Cancer Journal for Clinicians*, *74*(3), 229–263. https://doi.org/https://doi.org/10.3322/caac.21834

Burr, J. L., Johnson, K. C., Carmicheal, J. J., Lin, C., & Ganti, A. K. (2024). Combination Immunotherapy With Radiotherapy in Non-Small Cell Lung Cancer: A Review of Evidence. *Cancer Medicine*, *13*(21), e70402. https://doi.org/10.1002/cam4.70402

Byrne, H. M. (2010). Dissecting cancer through mathematics: from the cell to the animal model. *Nature Reviews Cancer*, *10*(3), 221–230. https://doi.org/10.1038/nrc2808

Cai, W., & Chen, X. (2008). Anti-Angiogenic Cancer Therapy Based on Integrin αvβ3 Antagonism. *Anti-Cancer Agents in Medicinal Chemistry*, *6*(5), 407–428. https://doi.org/10.2174/187152006778226530

Cao, J., Xu, Y., Zhang, J., Fang, T., Wu, F., Zhen, Y., Yu, X., Liu, Y., Li, J., & Dongkai Wang. (2024). "Nano-in-Micro" Structured Dry Powder Inhalers for pulmonary delivery: Advances and challenges. *Journal of Drug Delivery Science and Technology*, *96*, 105648. https://doi.org/https://doi.org/10.1016/j.jddst.2024.105648

Chang, J., Mancuso, M. R., Maier, C., Liang, X., Yuki, K., Yang, L., Kwong, J. W., Wang, J., Rao, V., Vallon, M., Kosinski, C., Zhang, J. J. H., Mah, A. T., Xu, L., Li, L., Gholamin, S., Reyes, T. F., Li, R., Kuhnert, F., … Kuo, C. J. (2017). Gpr124 is essential for blood–brain barrier integrity in central nervous system disease. *Nature Medicine*, *23*(4), 450–460. https://doi.org/10.1038/nm.4309

Chen, L., & Flies, D. B. (2013). Molecular mechanisms of T cell co-stimulation and co-inhibition. *Nature Reviews. Immunology*, *13*(4), 227–242. https://doi.org/10.1038/nri3405

Chen, M. Y., Zeng, Y. C., & Zhao, X. H. (2022). Chemotherapy- and Immune-Related Gene Panel in Prognosis Prediction and Immune Microenvironment of SCLC. *Frontiers in Cell and Developmental Biology*, *10*(June), 1–17. https://doi.org/10.3389/fcell.2022.893490

Chen, X., Yang, J., Wang, L., & Liu, B. (2020). Personalized neoantigen vaccination with synthetic long peptides: recent advances and future perspectives. *Theranostics*, *10*(13), 6011–6023. https://doi.org/10.7150/thno.38742

Cheng, X., & Chen, H. (2014). Tumor heterogeneity and resistance to EGFR-targeted therapy in advanced nonsmall cell lung cancer: challenges and perspectives. *OncoTargets and Therapy*, *7*, 1689–1704. https://doi.org/10.2147/OTT.S66502

Chittasupho, C., Angklomklew, J., Thongnopkoon, T., Senavongse, W., Jantrawut, P., & Ruksiriwanich, W.

(2021). Biopolymer Hydrogel Scaffolds Containing Doxorubicin as A Localized Drug Delivery System for Inhibiting Lung Cancer Cell Proliferation. In *Polymers* (Vol. 13, Issue 20, p. 3580). https://doi.org/10.3390/polym13203580

D.Q.A.T.E. Board, Non-Small Cell Lung Cancer Treatment (PDQ®): Health Professional Version, in: PDQ Cancer Information Summaries, National Cancer Institute (US), Bethesda (MD). (n.d.). *Md*.

Dearden, S., Stevens, J., & Blowers, D. (2013). *Mutation incidence and coincidence in non small-cell lung cancer : meta-analyses by ethnicity and histology*. *May*, 2371–2376. https://doi.org/10.1093/annonc/mdt205

Decker, S., Taschauer, A., Geppl, E., Pirhofer, V., Schauer, M., Pöschl, S., Kopp, F., Richter, L., Ecker, G. F., Sami, H., & Ogris, M. (2022). Structure-based peptide ligand design for improved epidermal growth factor receptor targeted gene delivery. *European Journal of Pharmaceutics and Biopharmaceutics*, *176*, 211–221. https://doi.org/https://doi.org/10.1016/j.ejpb.2022.05.004

Deng, Z., Siraj, A. A., Lowry, I., Ruan, E., Patel, R., Gao, W., Kalin, T. V, & Kalinichenko, V. V. (2025). Nanoparticle-Based Delivery Systems for Synergistic Therapy in Lung Cancers. In *Bioengineering* (Vol. 12, Issue 9, p. 968). https://doi.org/10.3390/bioengineering12090968

Desale, K., Kuche, K., & Jain, S. (2021). Cell-penetrating peptides (CPPs): An overview of applications for improving the potential of nanotherapeutics. *Biomaterials Science*, *9*(4), 1153–1188. https://doi.org/10.1039/d0bm01755h

Díaz-Perlas, C., & Oller-Salvia, B. (2023). Chemically Enhanced Peptide and Protein Therapeutics. In *Pharmaceutics* (Vol. 15, Issue 3, p. 827). https://doi.org/10.3390/pharmaceutics15030827

Doebele, R. C. (2019). Acquired Resistance Is Oncogene and Drug Agnostic. *Cancer Cell*, *36*(4), 347–349. https://doi.org/10.1016/j.ccell.2019.09.011

Drakopanagiotakis, F., Krauss, E., Michailidou, I., Drosos, V., Anevlavis, S., Günther, A., & Steiropoulos, P. (2024). Lung Cancer and Interstitial Lung Diseases. *Cancers*, *16*(16). https://doi.org/10.3390/cancers16162837

Duan, S., Lou, X., Chen, S., Jiang, H., Chen, D., Yin, R., Li, M., Gou, Y., Zhao, W., Sun, L., & Qian, F. (2023). Macrophage LMO7 deficiency facilitates inflammatory injury via metabolic-epigenetic

reprogramming. *Acta Pharmaceutica Sinica B*, *13*(12), 4785–4800. https://doi.org/https://doi.org/10.1016/j.apsb.2023.09.012

Eble, J. A., & Haier, J. (2006). Integrins in cancer treatment. *Current Cancer Drug Targets*, *6*(2), 89–105. https://doi.org/10.2174/156800906776056518

Esmaeili, A. (2025). Investigating the Impact of Peptide-Based Vaccines on Various Types of Cancer: A Systematic Review. *Cancer Investigation*, *43*(8), 657–680. https://doi.org/10.1080/07357907.2025.2568964

Fan, D.-Y., Tian, Y., & Liu, Z.-J. (2019). Injectable Hydrogels for Localized Cancer Therapy. *Frontiers in Chemistry*, *7*, 675. https://doi.org/10.3389/fchem.2019.00675

Fan, T., Yu, X., Shen, B., & Sun, L. (2017). Peptide Self-Assembled Nanostructures for Drug Delivery Applications. *Journal of Nanomaterials*, *2017*. https://doi.org/10.1155/2017/4562474

Fang, W., Li, X., Wang, Q., Meng, X., Zheng, W., Sun, L., Yao, W., Zhuang, W., Fan, Y., Zhuo, M., Luo, Y., Zhang, Z., Song, X., Yang, R., Yang, J., Jin, X., Diao, Y., Ge, J., & Zhang, L. (2025). Sacituzumab tirumotecan versus docetaxel for previously treated EGFR-mutated advanced non-small cell lung cancer: multicentre, open label, randomised controlled trial. *BMJ (Clinical Research Ed.)*, *389*, e085680. https://doi.org/10.1136/bmj-2025-085680

Fillingim. (2017). 乳鼠心肌提取 HHS Public Access. *Physiology & Behavior*, *176*(10), 139–148. https://doi.org/10.1016/j.addr.2016.06.015.Peptide

Friedlaender, A., Perol, M., Banna, G. L., Parikh, K., & Addeo, A. (2024). Oncogenic alterations in advanced NSCLC: a molecular super-highway. *Biomarker Research*, *12*(1), 24. https://doi.org/10.1186/s40364-024-00566-0

Genta, I., Chiesa, E., Colzani, B., Modena, T., Conti, B., & Dorati, R. (2017). GE11 Peptide as an Active Targeting Agent in Antitumor Therapy: A Minireview. *Pharmaceutics*, *10*(1). https://doi.org/10.3390/pharmaceutics10010002

Georgakopoulos, I., Kouloulias, V., Ntoumas, G., Desse, D., Koukourakis, I., Kougioumtzopoulou, A., Charpidou, A., Syrigos, K. N., & Zygogianni, A. (2024). Combined use of radiotherapy and tyrosine kinase inhibitors in the management of metastatic non-small cell lung cancer: A literature review.

Critical Reviews in Oncology/Hematology, *204*, 104520. https://doi.org/https://doi.org/10.1016/j.critrevonc.2024.104520

Goshisht, M. K., Goshisht, A., Bajpai, A., & Bajpai, A. (2025). Recent advances in biomedical applications of smart nanomaterials: a comprehensive review. *RSC Pharmaceutics*, *2*(6), 1227–1267. https://doi.org/10.1039/d5pm00137d

Guo, W., & Giancotti, F. G. (2004). Integrin signalling during tumour progression. *Nature Reviews. Molecular Cell Biology*, *5*(10), 816–826. https://doi.org/10.1038/nrm1490

Guy, P. M., Platko, J. V, Cantley, L. C., Cerione, R. A., & Carraway, K. L. (1994). Insect cell-expressed p180erbB3 possesses an impaired tyrosine kinase activity. *Proceedings of the National Academy of Sciences*, *91*(17), 8132–8136. https://doi.org/10.1073/pnas.91.17.8132

Hailing, T., Yonghong, P., Yufeng, Z., & Haitao, T. (2022). Challenges for the application of EGFR-targeting peptide GE11 in tumor diagnosis and treatment. *Journal of Controlled Release*, *349*, 592–605. https://doi.org/https://doi.org/10.1016/j.jconrel.2022.07.018

Han, H., Hou, Y., Chen, X., Zhang, P., Kang, M., Jin, Q., Ji, J., & Gao, M. (2020). Metformin-Induced Stromal Depletion to Enhance the Penetration of Gemcitabine-Loaded Magnetic Nanoparticles for Pancreatic Cancer Targeted Therapy. *Journal of the American Chemical Society*, *142*(10), 4944–4954. https://doi.org/10.1021/jacs.0c00650

Harada, H., Hiraoka, M., & Kizaka-Kondoh, S. (2002). Antitumor effect of TAT-oxygen-dependent degradation-caspase-3 fusion protein specifically stabilized and activated in hypoxic tumor cells. *Cancer Research*, *62*(7), 2013–2018.

Hassanein, S. S., Abdel-Mawgood, A. L., & Ibrahim, S. A. (2021). EGFR-Dependent Extracellular Matrix Protein Interactions Might Light a Candle in Cell Behavior of Non-Small Cell Lung Cancer. *Frontiers in Oncology*, *11*, 766659. https://doi.org/10.3389/fonc.2021.766659

Herbst, R. S. (2004). Review of epidermal growth factor receptor biology. *International Journal of Radiation Oncology, Biology, Physics*, *59*(2 Suppl), 21–26. https://doi.org/10.1016/j.ijrobp.2003.11.041

Hood, J. D., & Cheresh, D. A. (2002). Role of integrins in cell invasion and migration. *Nature Reviews. Cancer*, *2*(2), 91–100. https://doi.org/10.1038/nrc727

Hu, H., Chen, F., Dong, Y., Li, M., Xu, S., Qin, M., & Gong, P. (2020). Discovery of Novel c-Mesenchymal-Epithelia transition factor and histone deacetylase dual inhibitors. *European Journal of Medicinal Chemistry*, *204*, 112651. https://doi.org/https://doi.org/10.1016/j.ejmech.2020.112651

Hubbard, S. R. (1997). Crystal structure of the activated insulin receptor tyrosine kinase in complex with peptide substrate and ATP analog. *The EMBO Journal*, *16*(18), 5572–5581. https://doi.org/https://doi.org/10.1093/emboj/16.18.5572

Inamura, K. (2017). Lung cancer: understanding its molecular pathology and the 2015 wHO classification. *Frontiers in Oncology*, *7*(AUG), 1–7. https://doi.org/10.3389/fonc.2017.00193

Iyer, V. V. (2016). A Review of Stapled Peptides and Small Molecules to Inhibit Protein-Protein Interactions in Cancer. *Current Medicinal Chemistry*, *23*(27), 3025–3043. https://doi.org/10.2174/0929867323666160627103134

Izumchenko, E., Paz, K., Ciznadija, D., Sloma, I., Katz, A., Vasquez-Dunddel, D., Ben-Zvi, I., Stebbing, J., McGuire, W., Harris, W., Maki, R., Gaya, A., Bedi, A., Zacharoulis, S., Ravi, R., Wexler, L. H., Hoque, M. O., Rodriguez-Galindo, C., Pass, H., … Sidransky, D. (2017). Patient-derived xenografts effectively capture responses to oncology therapy in a heterogeneous cohort of patients with solid tumors. *Annals of Oncology*, *28*(10), 2595–2605. https://doi.org/https://doi.org/10.1093/annonc/mdx416

Jiang, T., Olson, E. S., Nguyen, Q. T., Roy, M., Jennings, P. A., & Tsien, R. Y. (2004). Tumor imaging by means of proteolytic activation of cell-penetrating peptides. *Proceedings of the National Academy of Sciences*, *101*(51), 17867–17872. https://doi.org/10.1073/pnas.0408191101

Jin, F., Lin, Y., Yuan, W., Wu, S., Yang, M., Ding, S., Liu, J., & Chen, Y. (2024). Recent advances in c-Met-based dual inhibitors in the treatment of cancers. *European Journal of Medicinal Chemistry*, *272*, 116477. https://doi.org/https://doi.org/10.1016/j.ejmech.2024.116477

Jiromaru, R., Nakagawa, T., & Yasumatsu, R. (2022). Advanced Nasopharyngeal Carcinoma: Current and Emerging Treatment Options. *Cancer Management and Research*, *14*, 2681–2689. https://doi.org/10.2147/CMAR.S341472

Karankar, V. S., Awasthi, S., & Srivastava, N. (2025). Peptide-driven strategies against lung cancer. *Life Sciences*, *366–367*, 123453. https://doi.org/https://doi.org/10.1016/j.lfs.2025.123453

Karati, D., Mukherjee, S., Prajapati, B., Bose, A., Paul, S., Elossaily, G. M., & Roy, S. (2024). A review on lipid-polymer hybrid nanocarriers in cancer. *Journal of Drug Delivery Science and Technology*, *97*, 105827. https://doi.org/https://doi.org/10.1016/j.jddst.2024.105827

Karnati, S., & Anusha, N. (2025). Peptidomimetics as next – generation therapeutic applications. *International Journal of Pharmacy and Pharmaceutical Science*, *7*(2), 407–412. https://doi.org/10.33545/26647222.2025.v7.i2e.234

Khan, M. S., Alqahtani, T., Al Shmrany, H., Gupta, G., Goh, K. W., Sahebkar, A., & Kesharwani, P. (2026). Enhanced permeability and retention (EPR) effect: Advances in nanomedicine for improved tumor targeting. *Biomaterials Advances*, *181*, 214636. https://doi.org/10.1016/j.bioadv.2025.214636

Kirn, D. H., & Thorne, S. H. (2009). Targeted and armed oncolytic poxviruses: a novel multi-mechanistic therapeutic class for cancer. *Nature Reviews Cancer*, *9*(1), 64–71. https://doi.org/10.1038/nrc2545

Kumar, C. (2005). Integrin αvβ3 as a Therapeutic Target for Blocking Tumor-Induced Angiogenesis. *Current Drug Targets*, *4*(2), 123–131. https://doi.org/10.2174/1389450033346830

Kumar, V. B. (2025). Recent advances in peptide-based self-assembled and metal coordinated nanocarriers for targeted cancer drug delivery. *European Journal of Pharmaceutics and Biopharmaceutics*, *217*(September), 114897. https://doi.org/10.1016/j.ejpb.2025.114897

Lahan, M., Saikia, T., Dutta, K., Baishya, R., Bharali, A., Baruah, S., Bharadwaj, R., Medhi, S., & Sahu, B. P. (2024). Multifunctional approach with LHRH-mediated PLGA nanoconjugate for site-specific codelivery of curcumin and BCL2 siRNA in mice lung cancer. *Future Journal of Pharmaceutical Sciences*, *10*(1), 163. https://doi.org/10.1186/s43094-024-00743-w

Lan, H., Zhang, Y., Dong, Z., & Zhang, X. (2025). Injectable hydrogel for controlled anlotinib delivery and enhanced anti-tumor therapy in non-small cell lung cancer. *Materials & Design*, *254*, 114103. https://doi.org/https://doi.org/10.1016/j.matdes.2025.114103

Laxio Arenas, J., Kaffy, J., & Ongeri, S. (2019). Peptides and peptidomimetics as inhibitors of protein-protein interactions involving β-sheet secondary structures. *Current Opinion in Chemical Biology*, *52*, 157–167. https://doi.org/10.1016/j.cbpa.2019.07.008

Li, C., Li, T., Tian, X., An, W., Wang, Z., Han, B., Tao, H., Wang, J., & Wang, X. (2024). Research

progress on the PEGylation of therapeutic proteins and peptides (TPPs). *Frontiers in Pharmacology*, *15*, 1353626. https://doi.org/10.3389/fphar.2024.1353626

Li, F., Deng, L., Jackson, K. R., Talukder, A. H., Katailiha, A. S., Bradley, S. D., Zou, Q., Chen, C., Huo, C., Chiu, Y., Stair, M., Feng, W., Bagaev, A., Kotlov, N., Svekolkin, V., Ataullakhanov, R., Miheecheva, N., Frenkel, F., Wang, Y., … Lizee, G. (2021). Neoantigen vaccination induces clinical and immunologic responses in non-small cell lung cancer patients harboring EGFR mutations. *Journal for Immunotherapy of Cancer*, *9*(7). https://doi.org/10.1136/jitc-2021-002531

Li, T., Lu, X.-M., Zhang, M.-R., Hu, K., & Li, Z. (2022). Peptide-based nanomaterials: Self-assembly, properties and applications. *Bioactive Materials*, *11*, 268–282. https://doi.org/https://doi.org/10.1016/j.bioactmat.2021.09.029

Li, Y., Wu, M., Fu, Y., Xue, J., Yuan, F., Qu, T., Rissanou, A. N., Wang, Y., Li, X., & Hu, H. (2024). Therapeutic stapled peptides: Efficacy and molecular targets. *Pharmacological Research*, *203*, 107137. https://doi.org/10.1016/j.phrs.2024.107137

Lian, F., Yang, H., Hong, R., Xu, H., Yu, T., Sun, G., Zheng, G., & Xie, B. (2024). Evaluation of the antitumor effect of neoantigen peptide vaccines derived from the translatome of lung cancer. *Cancer Immunology, Immunotherapy : CII*, *73*(7), 129. https://doi.org/10.1007/s00262-024-03670-0

Lindgren, M., Rosenthal-Aizman, K., Saar, K., Eiríksdóttir, E., Jiang, Y., Sassian, M., Östlund, P., Hällbrink, M., & Langel, Ü. (2006). Overcoming methotrexate resistance in breast cancer tumour cells by the use of a new cell-penetrating peptide. *Biochemical Pharmacology*, *71*(4), 416–425. https://doi.org/https://doi.org/10.1016/j.bcp.2005.10.048

Liu, L., Yang, L., Li, H., Shang, T., & Liu, L. (2026). The tumor microenvironment in lung cancer: Heterogeneity, therapeutic resistance and emerging treatment strategies (Review). *International Journal of Oncology*, *68*(1). https://doi.org/10.3892/ijo.2025.5824

Liu, S., Sun, X., Lu, H., Chen, D., Li, X., Li, L., Su, S., Zhao, Z., Cao, X., Liu, L., Lai, L., Lu, X., Shu, C., Wang, C., & Bai, C. (2023). Fullerene-based nanocomplex assists pulmonary delivery of siRNA for treating metastatic lung cancer. *Nano Today*, *50*, 101878. https://doi.org/https://doi.org/10.1016/j.nantod.2023.101878

Ma, W., Cheetham, A. G., & Cui, H. (2016). Building Nanostructures with Drugs. *Nano Today*, *11*(1), 13–

30. https://doi.org/10.1016/j.nantod.2015.11.003

Maity, S. K., Stahl, P., Hensel, A., Knauer, S., Hirschhäuser, C., & Schmuck, C. (2020). Cancer-Cell-Specific Drug Delivery by a Tumor-Homing CPP-Gossypol Conjugate Employing a Tracelessly Cleavable Linker. *Chemistry - A European Journal*, *26*(14), 3010–3015. https://doi.org/10.1002/chem.201905159

Mangal, S., Gao, W., Li, T., & Zhou, Q. T. (2017). Pulmonary delivery of nanoparticle chemotherapy for the treatment of lung cancers: challenges and opportunities. *Acta Pharmacologica Sinica*, *38*(6), 782–797. https://doi.org/10.1038/aps.2017.34

Mankoff, D. A., Farwell, M. D., Clark, A. S., & Pryma, D. A. (2017). Making Molecular Imaging a Clinical Tool for Precision Oncology: A Review. *JAMA Oncology*, *3*(5), 695–701. https://doi.org/10.1001/jamaoncol.2016.5084

Matsumura, Y., & Maeda, H. (1986). A new concept for macromolecular therapeutics in cancer chemotherapy: mechanism of tumoritropic accumulation of proteins and the antitumor agent smancs. *Cancer Research*, *46*(12 Pt 1), 6387–6392.

Milewska, S., Sadowska, A., Stefaniuk, N., Misztalewska-Turkowicz, I., Wilczewska, A. Z., Car, H., & Niemirowicz-Laskowska, K. (2024). Tumor-Homing Peptides as Crucial Component of Magnetic-Based Delivery Systems: Recent Developments and Pharmacoeconomical Perspective. *International Journal of Molecular Sciences*, *25*(11). https://doi.org/10.3390/ijms25116219

Milletti, F. (2012). Cell-penetrating peptides: classes, origin, and current landscape. *Drug Discovery Today*, *17*(15–16), 850–860. https://doi.org/10.1016/j.drudis.2012.03.002

Moiola, M., Memeo, M. G., & Quadrelli, P. (2019). Stapled Peptides—A Useful Improvement for Peptide-Based Drugs. In *Molecules* (Vol. 24, Issue 20, p. 3654). https://doi.org/10.3390/molecules24203654

Myrberg, H., Zhang, L., Mäe, M., & Langel, Ü. (2008). Design of a Tumor-Homing Cell-Penetrating Peptide. *Bioconjugate Chemistry*, *19*(1), 70–75. https://doi.org/10.1021/bc0701139

Naureen, F., Shah, Y., Rehman, M. U., Fazli Nasir, F. N., Pirzada, A. S., Al-Otaibi, J. S., Daglia, M., & Khan, H. (2025). Innovative inhalable dry powder: nanoparticles loaded with Crizotinib for targeted lung cancer therapy. *BMC Cancer*, *25*(1), 1526. https://doi.org/10.1186/s12885-025-15015-w

Nguyen, P. V., Hervé-Aubert, K., Chourpa, I., & Allard-Vannier, E. (2021). Active targeting strategy in nanomedicines using anti-EGFR ligands – A promising approach for cancer therapy and diagnosis. *International Journal of Pharmaceutics*, *609*, 121134. https://doi.org/https://doi.org/10.1016/j.ijpharm.2021.121134

Omidian, H., Cubeddu, L. X., & Wilson, R. L. (2025). Peptide-Functionalized Nanomedicine: Advancements in Drug Delivery, Diagnostics, and Biomedical Applications. In *Molecules* (Vol. 30, Issue 7, p. 1572). https://doi.org/10.3390/molecules30071572

Omidian, H., Nokhodchi, A., & Babanejad, N. (2025). Dry Powder Inhalers for Delivery of Synthetic Biomolecules. *Pharmaceuticals (Basel, Switzerland)*, *18*(2). https://doi.org/10.3390/ph18020175

Omori, S., Muramatsu, K., Kawata, T., Miyawaki, E., Miyawaki, T., Mamesaya, N., Kawamura, T., Kobayashi, H., Nakashima, K., Wakuda, K., Ono, A., Kenmotsu, H., Naito, T., Murakami, H., Sugino, T., & Takahashi, T. (2022). Trophoblast cell-surface antigen 2 expression in lung cancer patients and the effects of anti-cancer treatments. *Journal of Cancer Research and Clinical Oncology*, *148*(9), 2455–2463. https://doi.org/10.1007/s00432-021-03784-3

Passaro, A., Jänne, P. A., & Peters, S. (2023). Antibody-Drug Conjugates in Lung Cancer: Recent Advances and Implementing Strategies. *Journal of Clinical Oncology : Official Journal of the American Society of Clinical Oncology*, *41*(21), 3747–3761. https://doi.org/10.1200/JCO.23.00013

Patel, J. N., Ersek, J. L., & Kim, E. S. (2015). Lung cancer biomarkers, targeted therapies and clinical assays. *Translational Lung Cancer Research*, *4*(5), 503–514. https://doi.org/10.3978/j.issn.2218-6751.2015.06.02

Peng, S., Wang, W., Zhang, R., Wu, C., Pan, X., & Huang, Z. (2024). Nano-Formulations for Pulmonary Delivery: Past, Present, and Future Perspectives. In *Pharmaceutics* (Vol. 16, Issue 2, p. 161). https://doi.org/10.3390/pharmaceutics16020161

Pikor, L. A., Ramnarine, V. R., Lam, S., & Lam, W. L. (2013). Genetic alterations defining NSCLC subtypes and their therapeutic implications. *Lung Cancer*, *82*(2), 179–189. https://doi.org/https://doi.org/10.1016/j.lungcan.2013.07.025

Pouso, M. R., Melo, B. L., Gonçalves, J. J., Louro, R. O., Mendonça, A. G., Correia, I. J., & de Melo-Diogo, D. (2025). Injectable and implantable hydrogels for localized delivery of drugs and nanomaterials for

cancer chemotherapy: A review. *International Journal of Pharmaceutics*, *677*, 125640. https://doi.org/https://doi.org/10.1016/j.ijpharm.2025.125640

Prabhu, N., Rajinikanth, V., & Narayanan, M. (2026). Neoantigen-driven cancer vaccines in personalized oncology: progress, obstacles, and translational prospects. *Molecular Biology Reports*, *53*(1), 256. https://doi.org/10.1007/s11033-025-11421-2

Prior, I. A., Hood, F. E., & Hartley, J. L. (2020a). The frequency of ras mutations in cancer. *Cancer Research*, *80*(14), 2669–2974. https://doi.org/10.1158/0008-5472.CAN-19-3682

Prior, I. A., Hood, F. E., & Hartley, J. L. (2020b). *The Frequency of Ras Mutations in Cancer Ras Patterns across Datasets*. 2969–2974. https://doi.org/10.1158/0008-5472.CAN-19-3682

Qi, H., Li, F., Peng, L., Lin, J., & Jiang, Z. (2026). The progress of biomarkers detection to lung cancer. *Discover Oncology*. https://doi.org/10.1007/s12672-026-04625-w

Qu, N., & Song, K. (2024). *Albumin Nanoparticle-Based Drug Delivery Systems*. *July*.

Ramos-Paradas, J., Gómez-Sánchez, D., Rosado, A., Ucero, A. C., Ferrer, I., García-Luján, R., Zugazagoitia, J., Carrizo, N., Enguita, A. B., Conde, E., Garrido-Martin, E. M., & Paz-Ares, L. (2022). Comprehensive Characterization of Human Lung Large Cell Carcinoma Identifies Transcriptomic Signatures with Potential Implications in Response to Immunotherapy. *Journal of Clinical Medicine*, *11*(6), 1–20. https://doi.org/10.3390/jcm11061500

Rashid, H., Acharya, S., Muhammad, H., & Aziz, M. A. (2025). A comprehensive review of using nanomaterials in cancer immunotherapy: Pros and Cons of clinical usage. *3 Biotech*, *15*(7), 205. https://doi.org/10.1007/s13205-025-04362-x

Raso, M. G., Bota-Rabassedas, N., & Wistuba, I. I. (2021). Pathology and classification of SCLC. *Cancers*, *13*(4), 1–11. https://doi.org/10.3390/cancers13040820

Rathnayake, K., Patel, U., Hunt, E. C., & Singh, N. (2023). Fabrication of a Dual-Targeted Liposome-Coated Mesoporous Silica Core–Shell Nanoassembly for Targeted Cancer Therapy. *ACS Omega*, *8*(38), 34481–34498. https://doi.org/10.1021/acsomega.3c02901

Rauf, M. A., Rao, A., Sivasoorian, S. S., & Iyer, A. K. (2025). Nanotechnology-Based Delivery of CRISPR/Cas9 for Cancer Treatment: A Comprehensive Review. *Cells*, *14*(15).

https://doi.org/10.3390/cells14151136

Rodrigues Toledo, C., Tantawy, A. A., Lima Fuscaldi, L., Malavolta, L., & de Aguiar Ferreira, C. (2024). EGFR- and Integrin αVβ3-Targeting Peptides as Potential Radiometal-Labeled Radiopharmaceuticals for Cancer Theranostics. *International Journal of Molecular Sciences*, *25*(15). https://doi.org/10.3390/ijms25158553

Ruoslahti, E., Duza, T., & Zhang, L. (2005). Vascular Homing Peptides with Cell-Penetrating Properties. *Current Pharmaceutical Design*, *11*(28), 3655–3660. https://doi.org/10.2174/138161205774580787

Ruoslahti, E., & Pierschbacher, M. D. (1986). Arg-Gly-Asp: A versatile cell recognition signal. *Cell*, *44*(4), 517–518. https://doi.org/10.1016/0092-8674(86)90259-X

Sabir, F., Qindeel, M., Zeeshan, M., Ain, Q. U., Rahdar, A., Barani, M., González, E., & Aboudzadeh, M. A. (2021). Onco-receptors targeting in lung cancer via application of surface-modified and hybrid nanoparticles: A cross-disciplinary review. *Processes*, *9*(4). https://doi.org/10.3390/pr9040621

Safdar, I., Mahmood, S., Abdulwahab, M. K., Mohd Noor, S., Ge, Y., & Mohamed Sofian, Z. (2025). Inhalable Nanomaterial Discoveries for Lung Cancer Therapy: A Review. In *Pharmaceutics* (Vol. 17, Issue 8, p. 996). https://doi.org/10.3390/pharmaceutics17080996

Sainz de Aja, J., Dost, A. F. M., & Kim, C. F. (2021). Alveolar progenitor cells and the origin of lung cancer. *Journal of Internal Medicine*, *289*(5), 629–635. https://doi.org/https://doi.org/10.1111/joim.13201

Schnell, A., Bod, L., Madi, A., & Kuchroo, V. K. (2020). The yin and yang of co-inhibitory receptors: toward anti-tumor immunity without autoimmunity. *Cell Research*, *30*(4), 285–299. https://doi.org/10.1038/s41422-020-0277-x

Sen, T., Takahashi, N., Chakraborty, S., Takebe, N., Nassar, A. H., Karim, N. A., Puri, S., & Naqash, A. R. (2024). Emerging advances in defining the molecular and therapeutic landscape of small-cell lung cancer. *Nature Reviews. Clinical Oncology*, *21*(8), 610–627. https://doi.org/10.1038/s41571-024-00914-x

Seton-Rogers, S. (2016). Preventing competitive release. *Nature Reviews Cancer*, *16*(4), 199. https://doi.org/10.1038/nrc.2016.28

Shi, X., Deng, W., Pan, Y., Chen, Y., Wang, Y., & Wu, J. (2025). Overcoming resistance to immune checkpoint inhibitor in non-small cell lung cancer: the promise of combination therapy. *Frontiers in Immunology*, *16*, 1691980. https://doi.org/10.3389/fimmu.2025.1691980

Singh, N., Agrawal, S., Jiwnani, S., Khosla, D., Malik, P. S., Mohan, A., Penumadu, P., & Prasad, K. T. (2021). Lung Cancer in India. *Journal of Thoracic Oncology*, *16*(8), 1250–1266. https://doi.org/10.1016/j.jtho.2021.02.004

Soc, C., Cooper, B. M., Iegre, J., & Donovan, D. H. O. (2021). *Chem Soc Rev Peptides as a platform for targeted therapeutics for cancer : peptide – drug conjugates (PDCs)*. 1480–1494. https://doi.org/10.1039/d0cs00556h

Soler, M., González-Bártulos, M., Figueras, E., Massaguer, A., Feliu, L., Planas, M., Ribas, X., & Costas, M. (2016). Delivering aminopyridine ligands into cancer cells through conjugation to the cell-penetrating peptide BP1611Electronic supplementary information (ESI) available: Materials and methods; cell lines; synthesis of the metal binding peptides BP343, BP344, BP3. *Organic & Biomolecular Chemistry*, *14*(17), 4061–4070. https://doi.org/https://doi.org/10.1039/c6ob00470a

Stephens, A. J., Burgess-Brown, N. A., & Jiang, S. (2021). Beyond Just Peptide Antigens: The Complex World of Peptide-Based Cancer Vaccines. *Frontiers in Immunology*, *12*, 696791. https://doi.org/10.3389/fimmu.2021.696791

Su, H., Koo, J. M., & Cui, H. (2015). One-component nanomedicine. *Journal of Controlled Release : Official Journal of the Controlled Release Society*, *219*, 383–395. https://doi.org/10.1016/j.jconrel.2015.09.056

Sun, K., Qiu, J., Wang, Y., Han, F., Song, H., Wu, T., Wang, Y., & Jiao, W. (2025). Recent advances in hydrogel-based therapeutics for lung cancer. *RSC Advances*, *15*(55), 47676–47699. https://doi.org/10.1039/d5ra06157a

Suzuki, H., Fukuhara, M., Yamaura, T., Mutoh, S., Okabe, N., Yaginuma, H., Hasegawa, T., Yonechi, A., Osugi, J., Hoshino, M., Kimura, T., Higuchi, M., Shio, Y., Ise, K., Takeda, K., & Gotoh, M. (2013). Multiple therapeutic peptide vaccines consisting of combined novel cancer testis antigens and anti-angiogenic peptides for patients with non-small cell lung cancer. *Journal of Translational Medicine*, *11*, 97. https://doi.org/10.1186/1479-5876-11-97

Teesalu, T., Sugahara, K. N., & Ruoslahti, E. (2013). Tumor-penetrating peptides. *Frontiers in Oncology*, *3*, 216. https://doi.org/10.3389/fonc.2013.00216

Vagena, I.-A., Malapani, C., Gatou, M.-A., Lagopati, N., & Pavlatou, E. A. (2025). Enhancement of EPR Effect for Passive Tumor Targeting: Current Status and Future Perspectives. In *Applied Sciences* (Vol. 15, Issue 6, p. 3189). https://doi.org/10.3390/app15063189

Veluswamy, R., Mack, P. C., Houldsworth, J., Elkhouly, E., & Hirsch, F. R. (2021). REVIEW KRAS G12C e Mutant Non e Small Cell Lung Cancer Biology , Developmental Therapeutics , and Molecular Testing. *The Journal of Molecular Diagnostics*, *23*(5), 507–520. https://doi.org/10.1016/j.jmoldx.2021.02.002

Walensky, L. D., & Bird, G. H. (2014). *Hydrocarbon-Stapled Peptides: Principles, Practice, and Progress*.

Wang, Q., Zhang, X., Wei, W., & Cao, M. (2022). PET Imaging of Lung Cancers in Precision Medicine: Current Landscape and Future Perspective. *Molecular Pharmaceutics*, *19*(10), 3471–3483. https://doi.org/10.1021/acs.molpharmaceut.2c00353

Wang, W., Zhong, Z., Huang, Z., Hiew, T. N., Huang, Y., Wu, C., & Pan, X. (2024). *Nanomedicines for targeted pulmonary delivery : receptor-mediated strategy and alternatives*. 2820–2833. https://doi.org/10.1039/d3nr05487j

Wang, X., Niu, Y., & Bian, F. (2025). The progress of tumor vaccines clinical trials in non-small cell lung cancer. *Clinical & Translational Oncology : Official Publication of the Federation of Spanish Oncology Societies and of the National Cancer Institute of Mexico*, *27*(3), 1062–1074. https://doi.org/10.1007/s12094-024-03678-z

Wei, J., Li, W., Zhang, P., Guo, F., & Liu, M. (2024). Current trends in sensitizing immune checkpoint inhibitors for cancer treatment. *Molecular Cancer*, *23*(1), 279. https://doi.org/10.1186/s12943-024-02179-5

Werth, E. G., Roos, D., & Philip, E. T. (2024). Immunocapture LC-MS methods for pharmacokinetics of large molecule drugs. *Bioanalysis*, *16*(7), 165–177. https://doi.org/10.4155/bio-2023-0261

Xiao, X., Teng, F., Shi, C., Chen, J., Wu, S., Wang, B., Meng, X., Essiet Imeh, A., & Li, W. (2022). Polymeric nanoparticles-Promising carriers for cancer therapy. *Frontiers in Bioengineering and*

Biotechnology, *10*, 1024143. https://doi.org/10.3389/fbioe.2022.1024143

Xie, J., Li, J., Liu, Y., Wang, H., Wang, Y., Yang, Y., Chen, Y., Jiang, R., & Shao, R. (2023). Comparison of novel oncology drugs that received dual approval from the US accelerated approval and EU conditional marketing authorisation pathways, 2006-2021: a cross-sectional study. *BMJ Open*, *13*(6), e069132. https://doi.org/10.1136/bmjopen-2022-069132

Xiong, Q., Wen, H., Wang, J., Liu, Y., Wang, B., Zhao, S., Wang, Q., & Cheng, A. (2025). Applications and challenges of EGFR inhibitors in the treatment of nasopharyngeal carcinoma. *Biochemical and Biophysical Research Communications*, *788*, 152826. https://doi.org/https://doi.org/10.1016/j.bbrc.2025.152826

Yang, S., Li, P., Zhang, X., Liu, Z., Zhou, D., Chen, Y., Chen, Z., & Xiong, Y. (2025). High-Throughput Peptide Array Clinical Validation for Lung Adenocarcinoma Detection. *ACS Omega*, *10*(21), 21857–21866. https://doi.org/10.1021/acsomega.5c01737

Yang, Z., & Tam, K. Y. (2016). Anti-cancer synergy of dichloroacetate and EGFR tyrosine kinase inhibitors in NSCLC cell lines. *European Journal of Pharmacology*, *789*, 458–467. https://doi.org/10.1016/j.ejphar.2016.08.004

Yu, X., Fang, J., Xue, Y., Huang, Z., Wang, J., Zhu, X., Wang, X., Lv, T., & Wang, K. (2025). De novo discovery and preclinical evaluation of a novel peptide radiotracer for PET imaging of TROP-2 in non-small cell lung cancer. *Bioorganic Chemistry*, *167*, 109204. https://doi.org/https://doi.org/10.1016/j.bioorg.2025.109204

Zeinali, R., Zaeifi, D., Zolfaghari-Moghaddam, S. Y., Paul, M. K., & Biazar, E. (2025). Current Advances in Nanocarriers for Cancer Therapy. *International Journal of Nanomedicine*, *20*, 12217–12262. https://doi.org/10.2147/IJN.S548088

Zeng, F.-L., Xiao, Z., Wang, C.-Q., Jiang, Y., Shan, J.-L., Hu, S.-S., Huang, X.-R., Tang, Y.-H., Yao, X.-S., Zhang, T., Zeng, X.-T., Feng, J.-H., & Xiao, X. (2019). Clinical efficacy and safety of synthetic thymic peptides with chemotherapy for non-small cell lung cancer in China: A systematic review and meta-analysis of 27 randomized controlled trials following the PRISMA guidelines. *International Immunopharmacology*, *75*, 105747. https://doi.org/10.1016/j.intimp.2019.105747

Zeng, Y., Li, S., Zhang, S., Wang, L., Yuan, H., & Hu, F. (2022). Cell membrane coated-nanoparticles for

cancer immunotherapy. *Acta Pharmaceutica Sinica. B*, *12*(8), 3233–3254. https://doi.org/10.1016/j.apsb.2022.02.023

Zhang, C., Zhao, Y., Zhang, E., Jiang, M., Zhi, D., Chen, H., Cui, S., Zhen, Y., Cui, J., & Zhang, S. (2020). Co-delivery of paclitaxel and anti-VEGF siRNA by tripeptide lipid nanoparticle to enhance the anti-tumor activity for lung cancer therapy. *Drug Delivery*, *27*(1), 1397–1411. https://doi.org/10.1080/10717544.2020.1827085

Zhang, S.-W., Gong, C.-J., Su, M.-B., Chen, F., He, T., Zhang, Y.-M., Shen, Q.-Q., Su, Y., Ding, J., Li, J., Chen, Y., & Nan, F.-J. (2020). Synthesis and in Vitro and in Vivo Biological Evaluation of Tissue-Specific Bisthiazole Histone Deacetylase (HDAC) Inhibitors. *Journal of Medicinal Chemistry*, *63*(2), 804–815. https://doi.org/https://doi.org/10.1021/acs.jmedchem.9b01792

Zhang, X., Han, Y., Huang, W., Jin, M., & Gao, Z. (2021). The influence of the gut microbiota on the bioavailability of oral drugs. *Acta Pharmaceutica Sinica B*, *11*(7), 1789–1812. https://doi.org/https://doi.org/10.1016/j.apsb.2020.09.013

Zhang, X., Xiao, H., Na, F., Sun, J., Guan, Q., Liu, R., Cai, L., Li, H., & Zhao, M. (2025). Evolution of TROP2: Biological insights and clinical applications. *European Journal of Medicinal Chemistry*, *296*, 117863. https://doi.org/https://doi.org/10.1016/j.ejmech.2025.117863

Zhang, Z., Xia, Y., Wang, Z., Sun, Y., Pu, D., He, Y., Liu, R., Zhang, Y., Liu, Y., Yu, J., Ning, S., Feng, B., Wang, Y., & Wang, N. (2025). Ddx21 mutant peptide is an effective neoantigen in prophylactic lung cancer vaccines and activates long-term anti-tumor immunity. *Frontiers in Immunology*, *16*, 1500417. https://doi.org/10.3389/fimmu.2025.1500417

Zhao, Z., Wang, W., Wang, G., Huang, Z., Zhou, L., Lin, L., Ou, Y., Huang, W., Zhang, X., Wu, C., Tao, L., & Wang, Q. (2023). Dual peptides-modified cationic liposomes for enhanced Lung cancer gene therapy by a gap junction regulating strategy. *Journal of Nanobiotechnology*, *21*(1), 473. https://doi.org/10.1186/s12951-023-02242-1

Zhou, Y., Song, S., Yuan, B., Wu, Y., Gao, Y., Wan, G., & Li, G. (2022). A Novel CTLA-4 affinity peptide for cancer immunotherapy by increasing the integrin αvβ3 targeting. *Discover Oncology*, *13*(1), 99. https://doi.org/10.1007/s12672-022-00562-6

Zito Marino, F., Bianco, R., Accardo, M., Ronchi, A., Cozzolino, I., Morgillo, F., Rossi, G., & Franco, R.

(2019). Molecular heterogeneity in lung cancer: from mechanisms of origin to clinical implications. *International Journal of Medical Sciences*, *16*(7), 981–989. https://doi.org/10.7150/ijms.34739

Zorko, M., Jones, S., & Langel, Ü. (2022). Cell-penetrating peptides in protein mimicry and cancer therapeutics. *Advanced Drug Delivery Reviews*, *180*. https://doi.org/10.1016/j.addr.2021.114044

E-ISSN: 3139-0668
Volume: I, Issue: II